

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ **Correct!**

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

☐ $a + b = c$

```
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```

Project assets

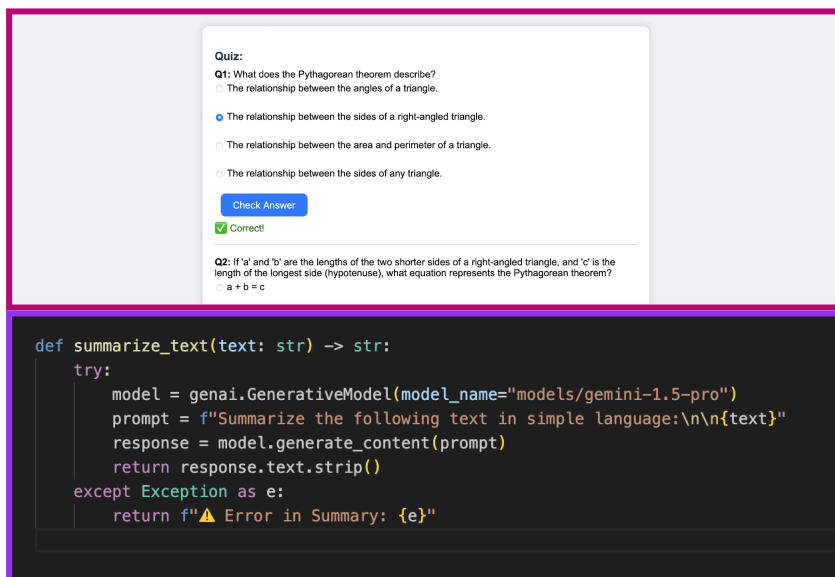
 File	 Description	 Status
 File		Not started ▾
 File		In progress ▾
 File		Under review ▾
 File		Approved ▾

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EduGenie: Google Gemini Powered Learning Assistant



Project Description

EduGenie is a lightweight AI-powered educational assistant that simplifies learning through generative AI. Designed for students of all academic levels, EduGenie enables users to:

- Ask questions and receive smart, concise answers
- Understand complex concepts through simplified explanations
- Generate quizzes from topics or text
- Receive personalized learning recommendations
- Summarize large educational passages

Built with **FastAPI** for the backend and a simple **HTML+CSS frontend**, EduGenie leverages lightweight and cloud-based AI models for local efficiency and cloud power. It works well on devices like the Mac M1, making it accessible to a broad range of learners and developers.

Scenarios

Scenario 1: A student wants to know about oceans and rivers uses EduGenie to ask "Which is the largest ocean?"

Scenario 2: A student wants to know the level of her understanding of "The Pythagoras Theorem" and clicks "Generate Quiz."

Scenario 3: A learner exploring SQL requests a learning p

~~path~~ath which is a structured plan with beginner to advanced topics, timelines, and suggestions.¶



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Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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```

⌌

⌌

import google.generativeai as genai⌌

⌌

def summarize_text(text: str) -> str:⌌

— try:⌌

— # Initialize the Gemini model⌌

— model =

genai.GenerativeModel(model_name="models/gemini-1.5-pro")⌌

— ⌌

— # Construct the prompt⌌

— prompt = f"Summarize the following text in simple language:\n\n{text}"⌌

— ⌌

— # Generate the response⌌

— response = model.generate_content(prompt)⌌

— ⌌

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~~return response.text.strip()~~

~~except Exception as e:~~

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**ARCHITECTURAL DIAGRAM** 

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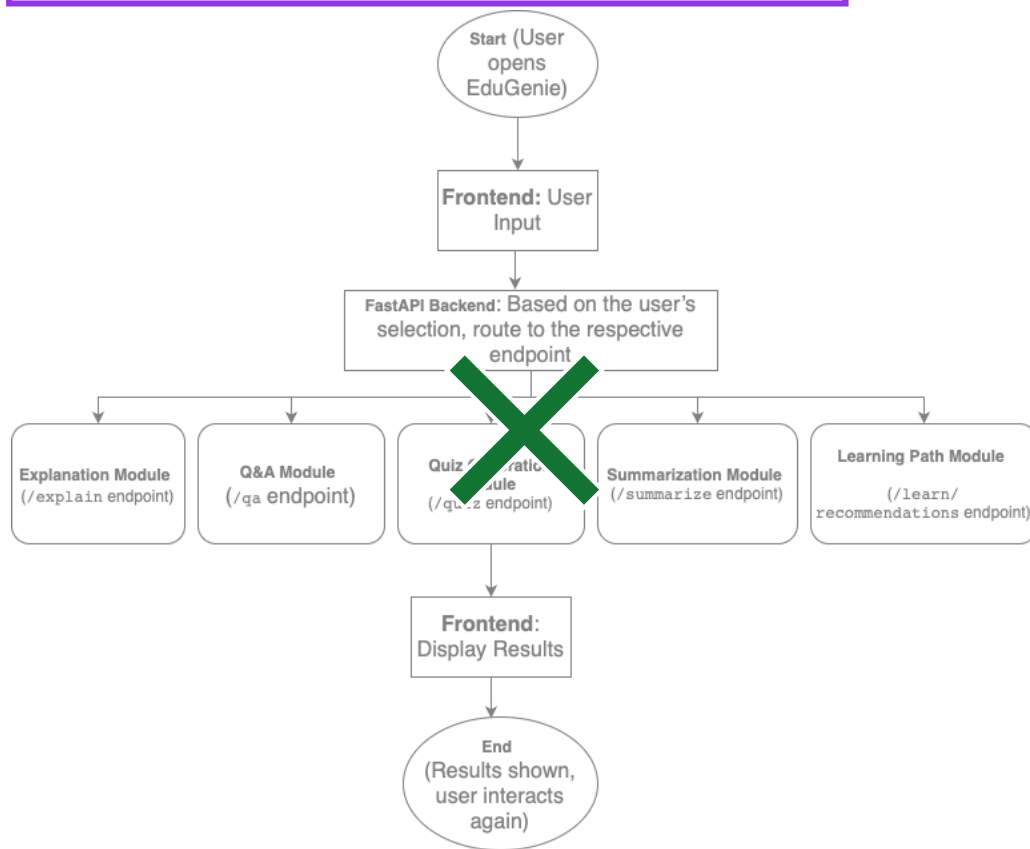
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Pre-requisites

Python 3.10+

1. 5. Click ****OK**** to save. Open a new terminal session to load the variable.

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```

2. `import google.generativeai as genai`
- 3.
4. `def summarize_text(text: str) -> str:`
5. `try:`
6. `model =`
`genai.GenerativeModel(model_name="models/gemini-1.5`
`-pro")`
7. `prompt = f"Summarize the following text in simple`
`language:\n\n{text}"`
8. `response = model.generate_content(prompt)`
9. `return response.text.strip()`
10. `except Exception as e:`
11. `return f"Error in Summary: {e}"`
- 12.
13. **### Option 2: Provide the API key explicitly in code**
- 14.
15. **You can pass the API key explicitly when initializing the client. Only do this**
16. **if you cannot use environment variables.**
- 17.

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18. ### Python

19.

20. from google import genai

21.

22. client = genai.Client(api_key="YOUR_API_KEY")

23.

24. interaction = client.interactions.create(

25. model="gemini-3.8-flash",

26. input="Explain how AI works in a few words"

27.)

28. print(interaction.output_text)

29. import google.generativeai as genai

30.

31. def summarize_text(text: str) -> str:

32. try:

33. model =

genai.GenerativeModel(model_name="models/gemini-1.5-pro")

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```

34. prompt = f"Summarize the following text in simple language:\n\n{text}"
35. response = model.generate_content(prompt)
36. return response.text.strip()
37. except Exception as e:
38. return f"Error in Summary: {e}"
- 39.
40. ### JavaScript
- 41.
42. import { GoogleGenAI } from "@google/genai";
- 43.
44. const ai = new GoogleGenAI({ apiKey: "YOUR_API_KEY" });
- 45.
46. async function main() {
47. const interaction = await ai.interactions.create({
48. model: "gemini-3.8-flash",
49. input: "Explain how AI works in a few words",
50. });

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        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

```
51. console.log(interaction.output_text);
52. }
53. async function main() {
54.   const interaction = await ai.interactions.create({
55.     model: "gemini-1.5-flash",
56.     input: "Explain how AI works in a few words",
57.   });
58.
59.   console.log(interaction.text);
60. }
61.
62. main();
63.
64. main();
65.
66. ### Java
67.
68. import com.google.genai.Client;
```

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        return response.text.strip()
    except Exception as e:
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```

69. `import`
`com.google.genai.gaos.models.interactions.CreateModelInteraction;`
70. `import`
`com.google.genai.gaos.models.interactions.Interaction;`
71. `import`
`com.google.genai.gaos.models.interactions.InteractionsInput;`
72. `import`
`com.google.genai.gaos.models.interactions.Model;`
73. `import`
`com.google.genai.gaos.models.operations.CreateInteractionRequestBody;`
- 74.
75. `Client client =`
`Client.builder().apiKey("YOUR_API_KEY").build();`
76. `import`
`com.google.genai.gaos.models.interactions.CreateModelInteraction;`
77. `import`
`com.google.genai.gaos.models.interactions.Interaction;`

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```

```
78. import
    com.google.genai.gaos.models.interactions.InteractionsIn
    put;

79. import com.google.genai.gaos.models.Model;

80.

81. public class GeminiExample {

82.     public static void main(String[] args) {

83.         // Code implementation for Google GenAI Interaction

84.         System.out.println("Gemini Interaction Initialized");

85.     }

86. }

87. import com.google.genai.gaos.models.Model;

88. import
    com.google.genai.gaos.models.interactions.CreateModelIn
    teraction;

89. import
    com.google.genai.gaos.models.interactions.Interaction;

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        return response.text.strip()
    except Exception as e:
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```

```
92. public class GeminiExample {
93.     public static void main(String[] args) {
94.
95.         // 1. Build interaction parameters
96.         CreateModelInteraction params =
            CreateModelInteraction.builder()
97.             .model(Model.of("gemini-1.5-flash"))
98.             .input(InteractionsInput.of("Explain how AI works
            in a few sentences."))
99.             .build();
100.
101.         // 2. Create interaction call
102.         Interaction interaction = client.interactions()
103.             .create(CreateModelInteractionRequestBody.of(params))
104.             .interaction()
105.             .get();
106.
107.         System.out.println("Gemini Interaction
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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

```
108.     }

109. }

110. import com.google.genai.gaos.models.Model;

111. import
    com.google.genai.gaos.models.interactions.CreateModelIn
    teraction;

112. import
    com.google.genai.gaos.models.interactions.Interaction;

113. import
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```

```
122.         .build();

123.

124.         // 2. Create interaction call

125.         Interaction interaction = client.interactions()

126.         .create(CreateModelInteractionRequestBody.of(params))

127.         .interaction()

128.         .get();

129.

130.         System.out.println("Gemini Interaction
        Initialized");

131.     }

132. }

133.

134.     CreateModelInteraction params =

135.     CreateModelInteraction.builder()

136.         .model(Model.of("gemini-3.8-flash"))

137.         .input(InteractionsInput.of("Explain how AI
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```
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141.     import
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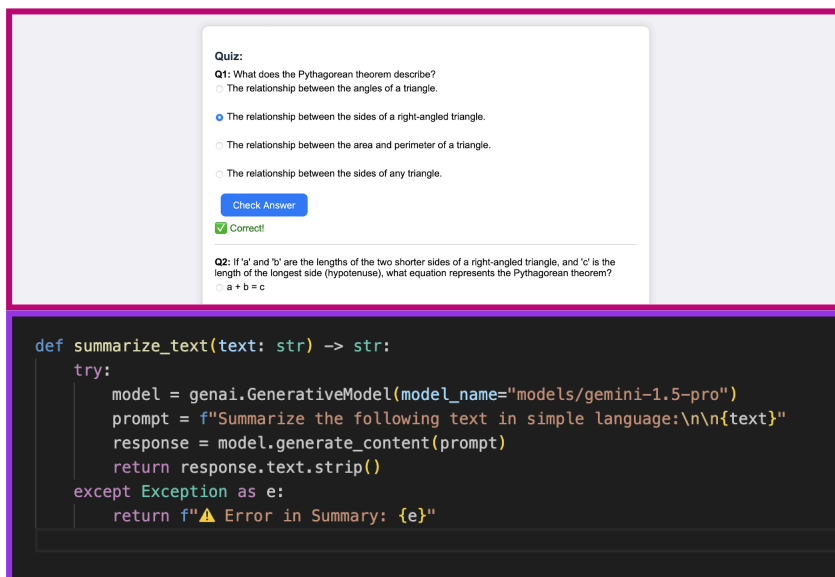
147.             // 1. Build interaction request body

148.             CreateModelInteractionRequestBody params =
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149.                 .input(InteractionsInput.of("Explain how AI
                    works in a few sentences."))

150.                 .build();

151.
```



```

152.         // 2. Create interaction call
153.         Interaction interaction = client.interactions()
154.             .create(params)
155.             .interaction()
156.             .get();
157.     import google.generativeai as genai
158.
159.     def summarize_text(text: str) -> str:
160.         try:
161.             model =
162.                 genai.GenerativeModel(model_name="models/gemini-1.5
163.                                     -pro")
164.
165.             prompt = f"Summarize the following text in simple
166.                     language:\n\n{text}"
167.
168.             response = model.generate_content(prompt)
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170.             return response.text.strip()
171.         except Exception as e:
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```
168.         // 3. Print output text safely
169.         System.out.println(interaction.text().orElse(""));
170.     }
171. }
172.
173.     Interaction interaction =
174.         client.interactions.create(CreateInteractionRequestBody.of
            (params)).interaction().get();
175.     curl
        "https://generativelanguage.googleapis.com/v1beta/inter
        actions" \
176.     -H "Content-Type: application/json" \
177.     -H "x-goog-api-key: YOUR_API_KEY" \
178.     -X POST \
179.     -d '{
180.         "model": "models/gemini-1.5-flash",
181.         "input": "Explain how AI works in a few sentences."
182.     }'
183.
```

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Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

☐ $a + b = c$

```
def summarize_text(text: str) -> str:
    try:
        model = genai.GenerativeModel(model_name="models/gemini-1.5-pro")
        prompt = f"Summarize the following text in simple language:\n\n{text}"
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

184.

`System.out.println(interaction.outputText().orElse(""));`

185.

186. `### REST`

187.

188. `curl`

`"https://generativelanguage.googleapis.com/v1beta/interactions" \`

189. `-H 'Content-Type: application/json' \`

190. `-H "x-goog-api-key: YOUR_API_KEY" \`

191. `-X POST \`

192. `-d '{`

193. `"model": "gemini-3.8-flash",`

194. `"input": "Explain how AI works in a few words"`

195. `}'`

196. `{`

197. `"model": "models/gemini-1.5-flash",`

198. `"input": "Explain how AI works in a few words"`

199. `}`

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

[Check Answer](#)

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Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

200.

201. **## Security and secret management**

202.

203. Treat your Gemini API key like a password. If compromised, others can consume

204. your project's quota, incur unexpected billing charges, and access private

205. resources.

206.

207. **### Critical security rules**

208.

209. - ****Keep keys confidential****: Never check API keys into source control systems like Git.

210. - ****Never expose keys client-side in production****: Do not hardcode API keys directly in web or mobile apps. Keys compiled in client-side code can be extracted by users. To secure client-side apps, run a backend proxy server to make the actual API calls.

211. {

212. "model": "models/gemini-1.5-flash",

213. "input": "Explain how AI works in a few words"

Quiz:

Q1: What does the Pythagorean theorem describe?

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

214. }

215.

216. ### Secret management best practices

217.

218. - **Environment variables**: Read keys from environment variables rather than configuration files.

219. - **Secret Manager** : For production, store your keys in a secure secret store such as [Google Cloud Secret Manager](https://cloud.google.com/secret-manager).

220. - **Billing alerts**: Set up billing alerts in the Google Cloud Console to notify you if usage or costs spike.

221.

222. ### Leak response checklist

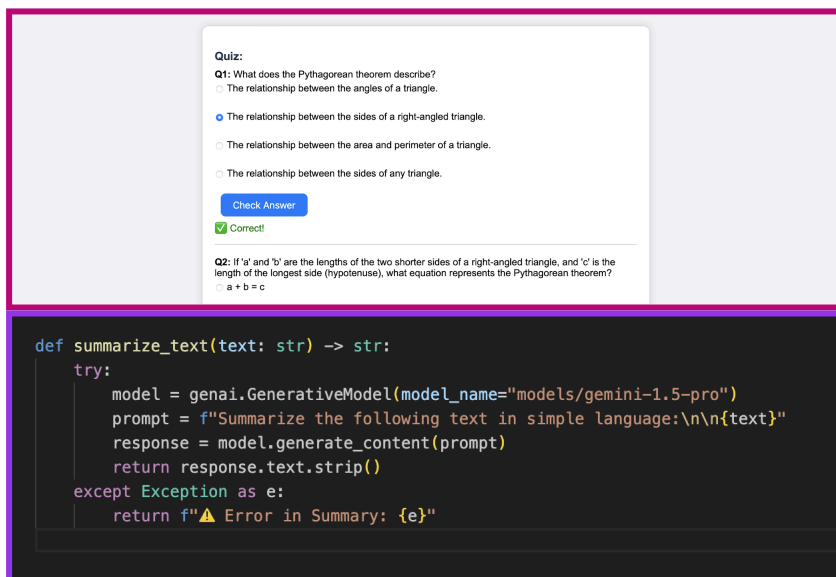
223.

224. If you suspect your API key has been leaked:

225.

226. 1. **Generate a new key**: Create a replacement key in Google AI Studio or the Cloud Console.

227. 2. **Update your application**: Deploy your code using the new key.



228. 3. ****Disable or delete the compromised key****: Disable the leaked key in the Cloud Console once the new key is verified. Do not delete the old key until the new key is fully active to avoid application downtime.

229. 4. ****Audit usage****: Check billing logs and API usage in the Google Cloud Console to identify unauthorized activity.

230. `import google.generativeai as genai`

231.

232. `def summarize_text(text: str) -> str:`

233. `try:`

234. `model =`

`genai.GenerativeModel(model_name="models/gemini-1.5-pro")`

235. `prompt = f"Summarize the following text in simple language:\n\n{text}"`

236. `response = model.generate_content(prompt)`

237. `return response.text.strip()`

238. `except Exception as e:`

239. `return f"Error in Summary: {e}"`

240.

241. **## Restricting and securing your keys**

Quiz:

Q1: What does the Pythagorean theorem describe?

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Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

242.

243. Adding restrictions to your API keys minimizes the potential damage if a key is

244. compromised.

245. `import google.generativeai as genai`

246.

247. `def summarize_text(text: str) -> str:`

248. `try:`

249. `model =`

`genai.GenerativeModel(model_name="models/gemini-1.5-pro")`

250. `prompt = f"Summarize the following text in simple language:\n\n{text}"`

251. `response = model.generate_content(prompt)`

252. `return response.text.strip()`

253. `except Exception as e:`

254. `return f"Error in Summary: {e}"`

255.

256. `### Apply request origin restrictions`

257.

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

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    try:
        model = genai.GenerativeModel(model_name="models/gemini-1.5-pro")
        prompt = f"Summarize the following text in simple language:\n\n{text}"
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

258. Origin restrictions limit which IP addresses, websites, or applications can use
259. your key.
- 260.
261. 1. Go to the [Google Cloud Console Credentials page](https://console.cloud.google.com/apis/credentials).
262. 2. Select your project, and click the name of the API key you want to restrict.
263. 3. Under ****Application restrictions****, select ****IP addresses**** (or the appropriate restriction type for your environment).
264. 4. Specify the allowed IP addresses or ranges, then click ****Save****.
265. `import google.generativeai as genai`
- 266.
267. `def summarize_text(text: str) -> str:`
268. `try:`
269. `model =`
`genai.GenerativeModel(model_name="models/gemini-1.5-pro")`

Quiz:

Q1: What does the Pythagorean theorem describe?

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

270. `prompt = f"Summarize the following text in simple language:\n\n{text}"`

271. `response = model.generate_content(prompt)`

272. `return response.text.strip()`

273. `except Exception as e:`

274. `return f"Error in Summary: {e}"`

275.

276. `### Securing unrestricted standard API keys`

277.

278. To continue using the Gemini API, you must secure any unrestricted keys.

279.

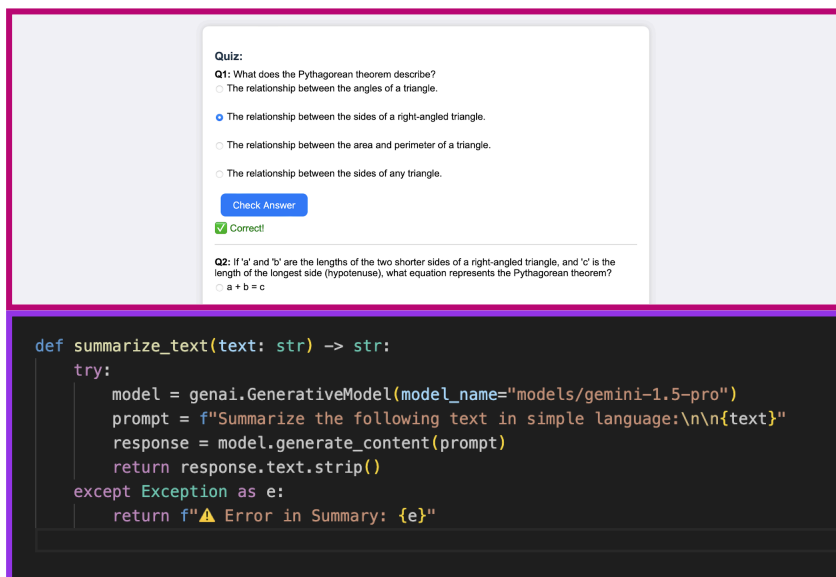
280. `#### Method A: Restrict the key to the Gemini API only (AI Studio)`

281. `import google.generativeai as genai`

282.

283. `def summarize_text(text: str) -> str:`

284. `try:`



285. model =
 genai.GenerativeModel(model_name="models/gemini-1.5-pro")
286. prompt = f"Summarize the following text in simple
 language:\n\n{text}"
287. response = model.generate_content(prompt)
288. return response.text.strip()
289. except Exception as e:
290. return f"Error in Summary: {e}"
- 291.
292. If you only use the key for the Gemini API, secure it
 directly in AI Studio:
- 293.
294. 1. On the ****API Keys**** page in [Google AI
 Studio](https://aistudio.google.com/api-keys), locate keys
 marked with the ****Unrestricted**** label.
295. 2. Hover over the label and click ****Add restrictions****
 in the dialog.
296. 3. Select ****Restrict to Gemini API only****.
297. 4. Click ****Restrict key**** to confirm.
298. import google.generativeai as genai

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

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☐ The relationship between the area and perimeter of a triangle.

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[Check Answer](#)

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

299.

300. `def summarize_text(text: str) -> str:`

301. `try:`

302. `model =`

`genai.GenerativeModel(model_name="models/gemini-1.5-pro")`

303. `prompt = f"Summarize the following text in simple language:\n\n{text}"`

304. `response = model.generate_content(prompt)`

305. `return response.text.strip()`

306. `except Exception as e:`

307. `return f"Error in Summary: {e}"`

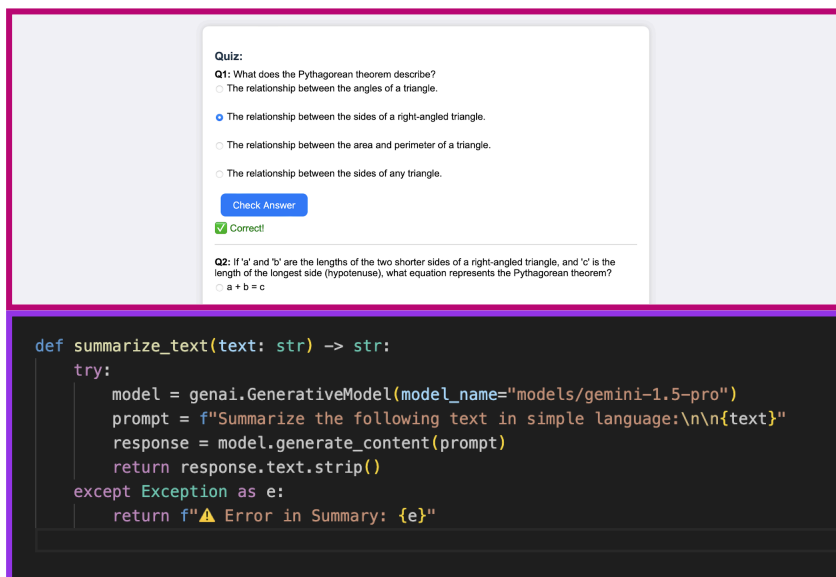
308.

309. `> [!NOTE]`

310. `> **Note:**` To restrict an API key, you must have the ``apikey.keys.update`` permission on the associated Google Cloud project. This permission is included in roles like `**API Keys Admin**` or `**Editor**`.

311.

312. `#### Method B: Restrict the key for other services`
(Google Cloud Console)



313.

314. If the key is shared with other Google APIs (not recommended), restrict it in

315. the Cloud Console. ****Note: Gemini API requests using this key will fail after**

316. these restrictions are applied. ******

317. `import google.generativeai as genai`

318.

319. `def summarize_text(text: str) -> str:`

320. `try:`

321. `model =
genai.GenerativeModel(model_name="models/gemini-1.5-pro")`

322. `prompt = f"Summarize the following text in simple language:\n\n{text}"`

323. `response = model.generate_content(prompt)`

324. `return response.text.strip()`

325. `except Exception as e:`

326. `return f"Error in Summary: {e}"`

327. `import google.generativeai as genai`

Quiz:

Q1: What does the Pythagorean theorem describe?

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[Check Answer](#)

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    try:
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        return response.text.strip()
    except Exception as e:
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```

328.

329. `def summarize_text(text: str) -> str:`

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331. `model =`
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332. `prompt = f"Summarize the following text in simple language:\n\n{text}"`

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334. `return response.text.strip()`

335. `except Exception as e:`

336. `return f"Error in Summary: {e}"`

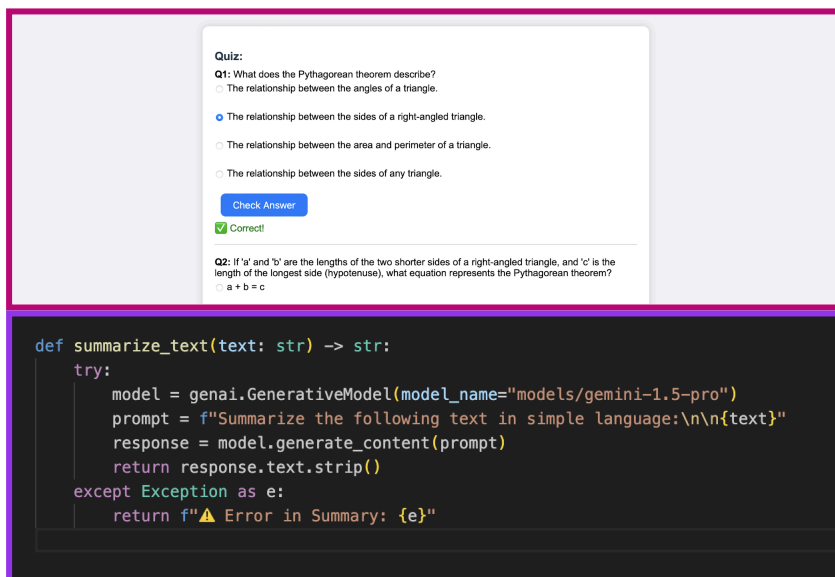
337. `import google.generativeai as genai`

338.

339. `def summarize_text(text: str) -> str:`

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341. `model =`
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342. prompt = f"Summarize the following text in simple
language:\n\n{text}"

343. response = model.generate_content(prompt)

344. return response.text.strip()

345. except Exception as e:

346. return f"Error in Summary: {e}"

347.

348. 1. Visit the [Google Cloud Console Credentials
page](https://console.cloud.google.com/apis/credentials).

349. 2. Select the project and the API key.

350. 3. Under ****API restrictions**** , use the ****Select API
restrictions**** drop-down to select the APIs you want this
key to access. Do not select the ****Generative**

351. Language API**.

352. 4. Click ****Save****. Create a separate, restricted key in
AI Studio to continue using the Gemini API.

353.

354. **### Blocked dormant keys**

355.

Quiz:

Q1: What does the Pythagorean theorem describe?

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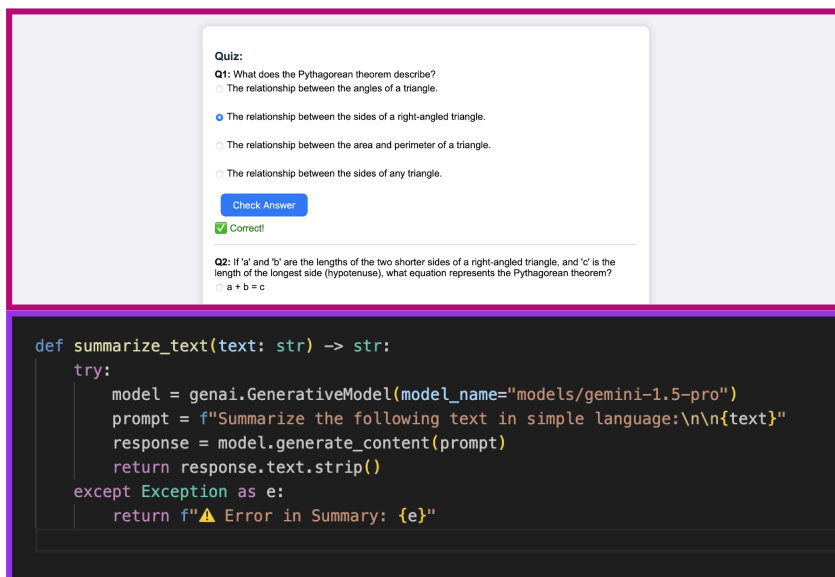
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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

356. Starting May 7, 2026, the Gemini API blocks unrestricted API keys that have
357. been dormant for an extended period. These keys show a ****Blocked**** tag in AI
358. Studio. You must generate a new key or use an existing restricted key to
359. continue.
- 360.
361. **##** Migrate to an auth key
- 362.
363. Follow these steps to create a new auth API key and update your applications:
- 364.
365. 1. Go to the [AI Studio API Keys page](https://aistudio.google.com/api-keys).
366. 2. Check the ****Key Type**** column to identify any keys listed as ****Standard****.
367. 3. Click ****Create API key**** to generate a new key. All new keys created in AI Studio are automatically created as auth keys.
368. 4. Copy the new auth API key.



369. 5. Update your application code, environment variables, and any deployment configurations to use the new auth API key.
370. 6. Test your application to confirm it works correctly with the new key.
371. 7. Once verified, delete or revoke your old traffic key to prevent misuse.
- 372.
373. > [!NOTE]
374. > **Note:** If you don't see your keys, you may need to import your Cloud projects into AI Studio. Refer to the [import projects](https://ai.google.dev/gemini-api/docs/api-key#import-projects) section for instructions.
- 375.
376. ## Limitations
- 377.
378. Google AI Studio imposes the following project and key management limitations:
- 379.
380. - You can create a maximum of 10 projects at a time from the Google AI Studio **Projects** page.

Quiz:

Q1: What does the Pythagorean theorem describe?

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        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

381. - The ****API keys**** and ****Projects**** pages display a maximum of 100 keys and 50 projects.
382. - Only API keys that are unrestricted or restricted specifically to the Generative Language API (Gemini API) are displayed.
- 383.
384. For advanced project management or to modify keys with other restrictions, use
385. the [Google Cloud Console credentials page](https://console.cloud.google.com/apis/credentials).
386. To use the Gemini API, you must authenticate your requests. You can
387. authenticate using a standard or authorization API key.
- 388.
389. [Create or view a Gemini API Key](https://aistudio.google.com/apikey)
- 390.
391. **## API key types: standard versus authorization**
- 392.
393. API keys provide access to the Gemini API, but their security characteristics

Quiz:

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    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

394. differ. The Gemini API is transitioning from standard API keys to authorization

395. keys to improve security:

396.

397. - **Standard API keys**: Associate requests with a Google Cloud project for billing and quota purposes. Standard keys don't identify a caller, which limits the granularity of permissions and access control they can support.

398. - **Authorization (auth) keys**: Bound directly to a Google Cloud service account. When you use an authorization key, your requests are processed under the identity of that bound service account, enabling granular access control. Authorization keys are restricted to the Generative Language API (Gemini API) by default and provide fast-acting leaked key enforcement that quickly stops the usage of leaked keys detected by our systems.

399.

400. To ensure secure usage, Gemini API will move from standard keys to auth keys:

401.

Quiz:

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        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

402. - ****Auth keys default****: Starting May 28, 2026, all new API keys created in Google AI Studio are automatically created as auth keys.

403. - ****Unrestricted keys rejected**** : The Gemini API rejects requests from ****unrestricted standard keys****. Standard API keys that have explicit restrictions applied continue to work. This restriction prevents the unauthorized use of keys that might be shared publicly or linked to other services.

404. - ****On September 2026**** : the Gemini API will reject requests from ****standard**

405. **keys**** . You must [migrate to auth keys](https://ai.google.dev/gemini-api/docs/api-key#migrate-to-auth-key) before this date to avoid service interruption. Make sure to migrate to auth keys before September 2026.

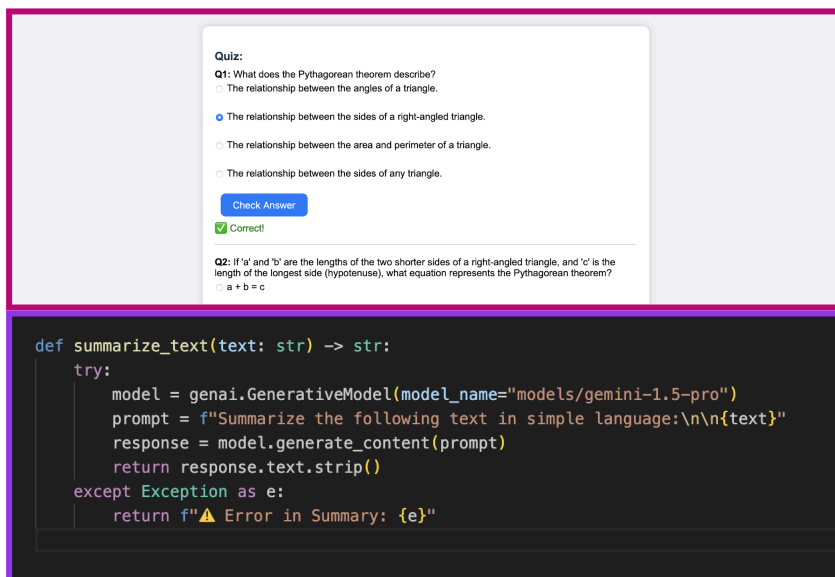
406.

407. > [!NOTE]

408. > ****Note:**** Requests authenticated by authorization keys are not recorded in Google Cloud service account usage metrics.

409.

410. ## Managing API keys in Google AI Studio



411.

412. You can manage your projects and keys directly in [Google AI Studio](<https://aistudio.google.com/apikey>).

413.

414. ### Google Cloud projects

415.

416. Every Gemini API key is associated with a [Google Cloud project](<https://cloud.google.com/resource-manager/docs/creating-managing-projects>).

417. Google Cloud projects manage billing, collaborators, and permissions. Google AI

418. Studio provides a lightweight interface to access these projects.

419.

420. - **Default project** : If you are a new user, Google AI Studio automatically creates a default Google Cloud project and API key after you accept the Terms of Service. You can rename this project by navigating to the **Projects** view in your dashboard.

Quiz:

Q1: What does the Pythagorean theorem describe?

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        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

421. - ****Existing projects****: If you already have a Google Cloud account, AI Studio does not create a default project. Instead, you must import your existing projects.

422.

423. **### Importing projects**

424.

425. By default, Google AI Studio does not display all of your Google Cloud

426. projects. You must import the projects you want to use:

427.

428. 1. Go to [Google AI Studio](https://aistudio.google.com).

429. 2. Open the ****Dashboard**** from the left panel and select ****Projects****.

430. 3. Click the ****Import projects**** button.

431. 4. Search for and select the Google Cloud project you want to import, then click ****Import****.

432. 5. Once imported, navigate to the ****API Keys**** page in the dashboard to create a key in that project.

433.

Quiz:

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Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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434. **### Troubleshooting key creation permissions**
- 435.
436. If the ****Create API key**** button is unavailable and displays the message:
437. **"You do not have permission to create a key in this project"**, you lack the
438. required IAM permissions.
- 439.
440. Ask your Google Cloud project or organization administrator to grant you a role
441. containing the following permissions (such as Project Editor):
- 442.
443. - ``resourcemanager.projects.get``: Allows AI Studio to verify the project.
444. - ``apikeys.keys.create``: Allows key generation.
445. - ``serviceusage.services.enable``: Ensures the Generative Language API is enabled.
446. - ``iam.serviceAccounts.create``: Required to create the linked service account.

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447. - `iam.serviceAccountApiKeyBindings.create`: Binds the service account to the API key.

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452. ## Setting up your environment

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454. Once you have a key, configure your environment to use it securely in your

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457. ### Option 1: Use environment variables (recommended)

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459. Set the environment variable `GEMINI\_API\_KEY` or `GOOGLE\_API\_KEY`. The Gemini

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461. set, `GOOGLE\_API\_KEY` takes precedence.

462.

463. Select your operating system to set the variable:

464.

465. `### Linux/macOS - Bash`

466.

467. Verify if you have a bash configuration file:

468.

469. `~/.bashrc`

470.

471. If not, create one and open it:

472.

473. `touch ~/.bashrc && open ~/.bashrc`

474.

475. Add the export command at the end of the file:

476.

477. `export GEMINI_API_KEY=<YOUR_API_KEY_HERE>`

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479. Save the file, then apply the changes:
- 480.
481. `source ~/.bashrc`
- 482.
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- 484.
485. Verify if you have a zsh configuration file:
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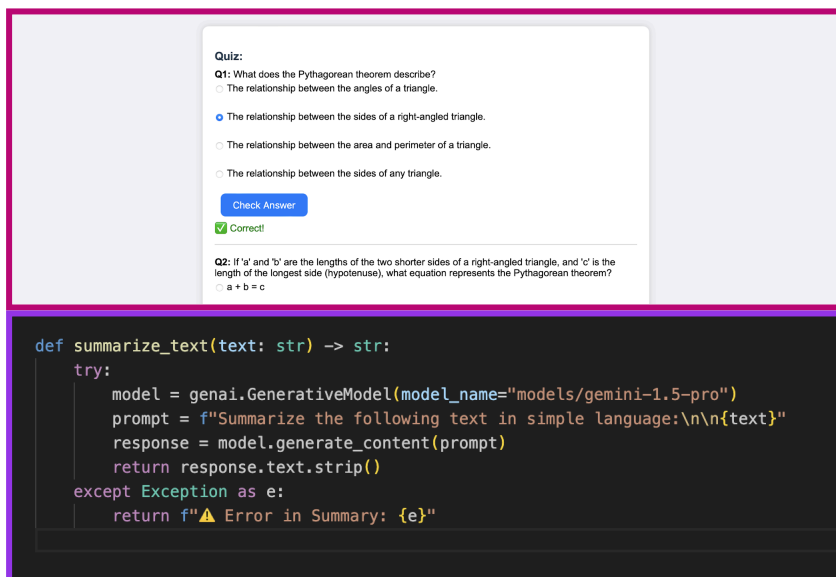
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- 500.
501. **### Windows**
- 502.
503. 1. Search for "Environment Variables" in the Windows search bar.
504. 2. Click ****Environment Variables**** in the System Properties dialog.
505. 3. Under ****User variables**** or ****System variables****, click ****New...****.
506. 4. Set the variable name to ``GEMINI_API_KEY`` and the value to your API key.
507. 5. Click ****OK**** to save. Open a new terminal session to load the variable.
- 508.
509. **### Option 2: Provide the API key explicitly in code**
- 510.



511. You can pass the API key explicitly when initializing the client. Only do this
512. if you cannot use environment variables.
- 513.
514. `### Python`
- 515.
516. `from google import genai`
- 517.
518. `client = genai.Client(api_key="YOUR_API_KEY")`
- 519.
520. `interaction = client.interactions.create(`
521. `model="gemini-3.8-flash",`
522. `input="Explain how AI works in a few words"`
523. `)`
524. `print(interaction.output_text)`
- 525.
526. `### JavaScript`
- 527.
528. `import { GoogleGenAI } from "@google/genai";`

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529.

530. const ai = new GoogleGenAI({ apiKey:
 "YOUR_API_KEY" });

531.

532. async function main() {

533. const interaction = await ai.interactions.create({

534. model: "gemini-3.8-flash",

535. input: "Explain how AI works in a few words",

536. });

537. console.log(interaction.output_text);

538. }

539.

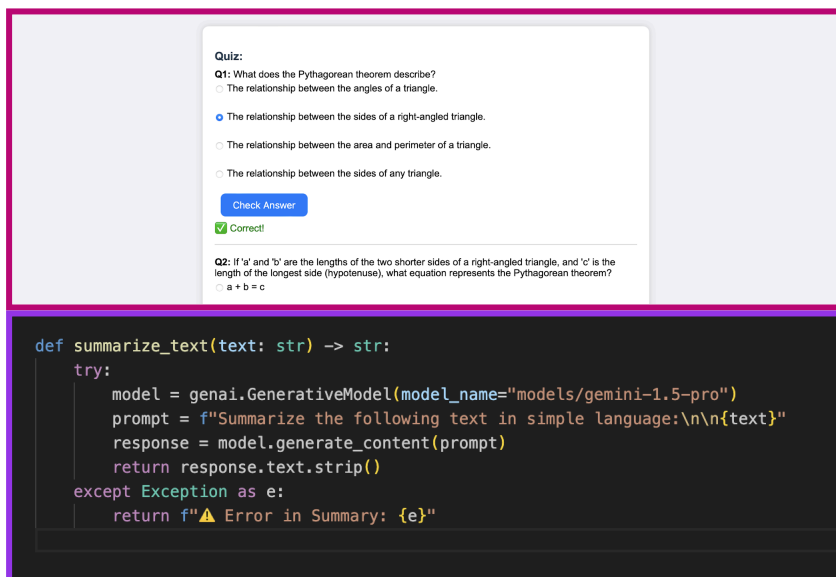
540. main();

541.

542. ### Java

543.

544. import com.google.genai.Client;



```
545.     import
        com.google.genai.gaos.models.interactions.CreateModelIn
        teraction;

546.     import
        com.google.genai.gaos.models.interactions.Interaction;

547.     import
        com.google.genai.gaos.models.interactions.InteractionsIn
        put;

548.     import
        com.google.genai.gaos.models.interactions.Model;

549.     import
        com.google.genai.gaos.models.operations.CreateInteracti
        onRequestBody;

550.

551.     Client client =
        Client.builder().apiKey("YOUR_API_KEY").build();

552.

553.     CreateModelInteraction params =

554.         CreateModelInteraction.builder()

555.             .model(Model.of("gemini-3.8-flash"))

556.             .input(InteractionsInput.of("Explain how AI
        works in a few sentences."))
```

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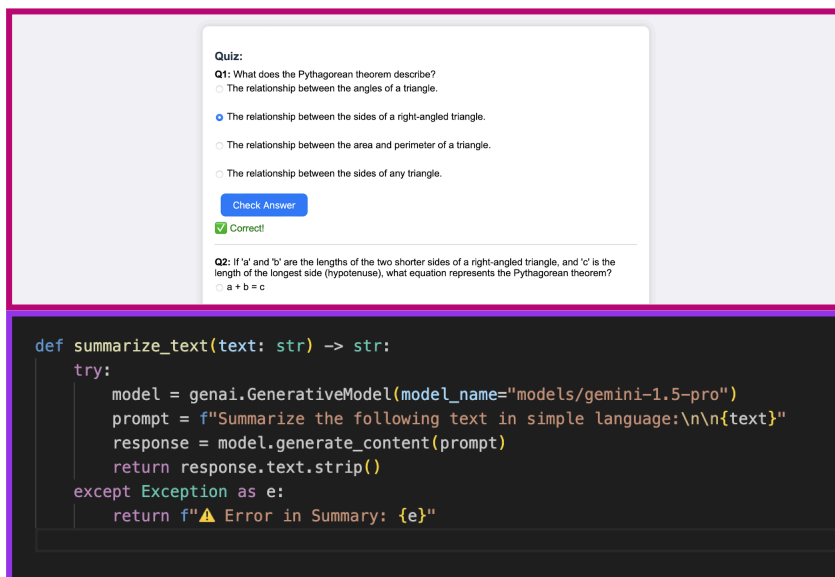
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557. .build();
- 558.
559. Interaction interaction =
560.
 client.interactions.create(CreateInteractionRequestBody.of
 (params)).interaction().get();
- 561.
562.
 System.out.println(interaction.outputText().orElse(""));
- 563.
564. ### RESTTo use the Gemini API, you must authenticate
 your requests. You can
565. authenticate using a standard or authorization API key.
- 566.
567. [Create or view a Gemini API
 Key](https://aistudio.google.com/apikey)
- 568.
569. ## API key types: standard versus authorization
- 570.



571. API keys provide access to the Gemini API, but their security characteristics
572. differ. The Gemini API is transitioning from standard API keys to authorization
573. keys to improve security:
- 574.
575. - **Standard API keys**: Associate requests with a Google Cloud project for billing and quota purposes. Standard keys don't identify a caller, which limits the granularity of permissions and access control they can support.
576. - **Authorization (auth) keys**: Bound directly to a Google Cloud service account. When you use an authorization key, your requests are processed under the identity of that bound service account, enabling granular access control. Authorization keys are restricted to the Generative Language API (Gemini API) by default and provide fast-acting leaked key enforcement that quickly stops the usage of leaked keys detected by our systems.
- 577.
578. To ensure secure usage, Gemini API will move from standard keys to auth keys:
- 579.

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580. - ****Auth keys default****: Starting May 28, 2026, all new API keys created in Google AI Studio are automatically created as auth keys.

581. - ****Unrestricted keys rejected**** : The Gemini API rejects requests from ****unrestricted standard keys****. Standard API keys that have explicit restrictions applied continue to work. This restriction prevents the unauthorized use of keys that might be shared publicly or linked to other services.

582. - ****On September 2026**** : the Gemini API will reject requests from ****standard**

583. **keys**** . You must [migrate to auth keys](https://ai.google.dev/gemini-api/docs/api-key#migrate-to-auth-key) before this date to avoid service interruption. Make sure to migrate to auth keys before September 2026.

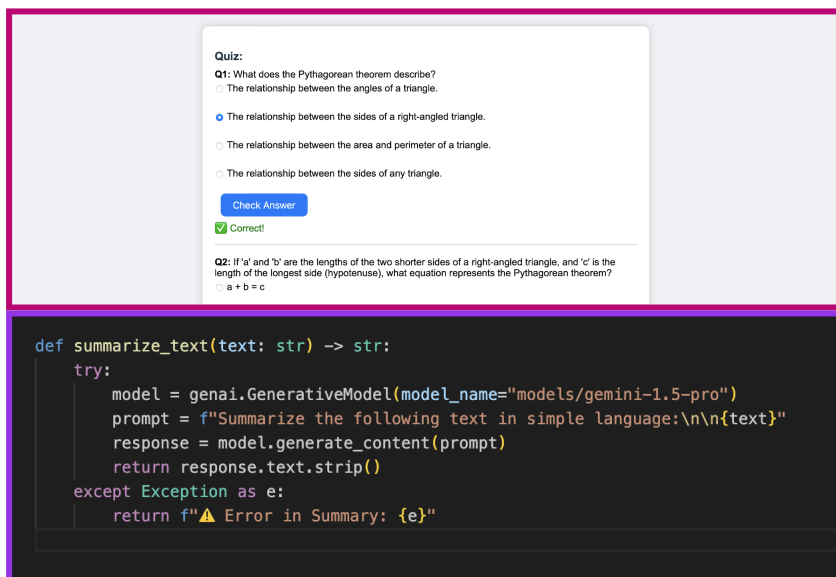
584.

585. > [!NOTE]

586. > ****Note:**** Requests authenticated by authorization keys are not recorded in Google Cloud service account usage metrics.

587.

588. **## Managing API keys in Google AI Studio**



589.

590. You can manage your projects and keys directly in [Google AI Studio](<https://aistudio.google.com/apikey>).

591.

592. ### Google Cloud projects

593.

594. Every Gemini API key is associated with a [Google Cloud project](<https://cloud.google.com/resource-manager/docs/creating-managing-projects>).

595. Google Cloud projects manage billing, collaborators, and permissions. Google AI

596. Studio provides a lightweight interface to access these projects.

597.

598. - **Default project** : If you are a new user, Google AI Studio automatically creates a default Google Cloud project and API key after you accept the Terms of Service. You can rename this project by navigating to the **Projects** view in your dashboard.

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599. - ****Existing projects****: If you already have a Google Cloud account, AI Studio does not create a default project. Instead, you must import your existing projects.

600.

601. **### Importing projects**

602.

603. By default, Google AI Studio does not display all of your Google Cloud

604. projects. You must import the projects you want to use:

605.

606. 1. Go to [Google AI Studio](https://aistudio.google.com).

607. 2. Open the ****Dashboard**** from the left panel and select ****Projects****.

608. 3. Click the ****Import projects**** button.

609. 4. Search for and select the Google Cloud project you want to import, then click ****Import****.

610. 5. Once imported, navigate to the ****API Keys**** page in the dashboard to create a key in that project.

611.

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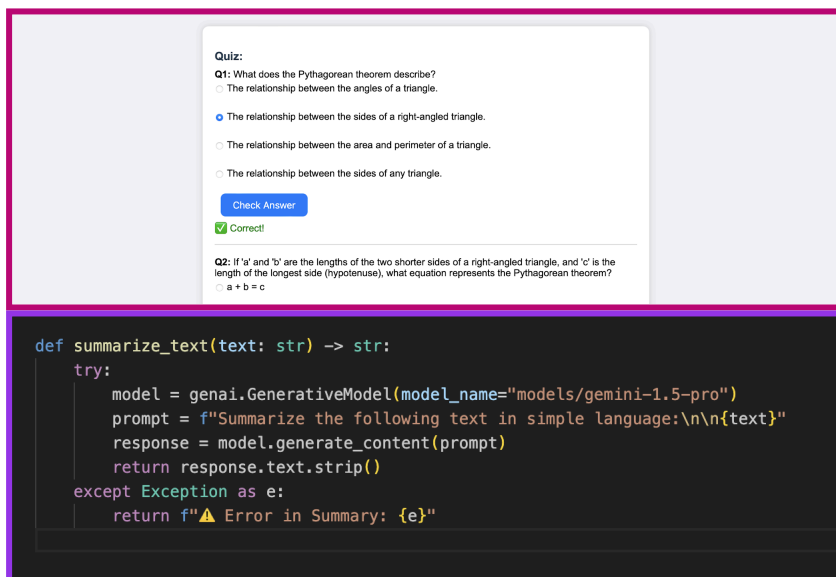
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☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

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[Check Answer](#)

✔ **Correct!**

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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        return response.text.strip()
    except Exception as e:
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```

707.

708. const ai = new GoogleGenAI({ apiKey:
 "YOUR_API_KEY" });

709.

710. async function main() {

711. const interaction = await ai.interactions.create({

712. model: "gemini-3.8-flash",

713. input: "Explain how AI works in a few words",

714. });

715. console.log(interaction.output_text);

716. }

717.

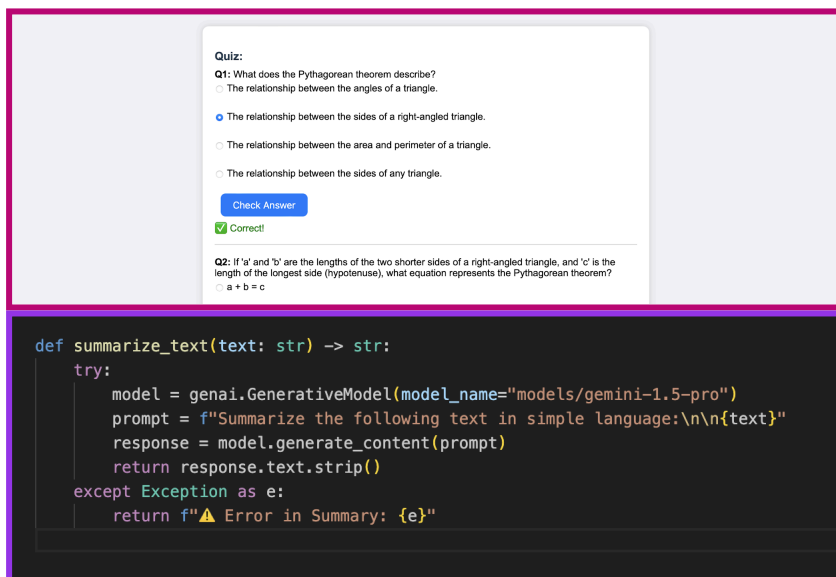
718. main();

719.

720. ### Java

721.

722. import com.google.genai.Client;



```
723.     import
        com.google.genai.gaos.models.interactions.CreateModelIn
        teraction;

724.     import
        com.google.genai.gaos.models.interactions.Interaction;

725.     import
        com.google.genai.gaos.models.interactions.InteractionsIn
        put;

726.     import
        com.google.genai.gaos.models.interactions.Model;

727.     import
        com.google.genai.gaos.models.operations.CreateInteracti
        onRequestBody;

728.

729.     Client client =
        Client.builder().apiKey("YOUR_API_KEY").build();

730.

731.     CreateModelInteraction params =

732.         CreateModelInteraction.builder()

733.             .model(Model.of("gemini-3.8-flash"))

734.             .input(InteractionsInput.of("Explain how AI
        works in a few sentences."))
```

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```

```
735.         .build();

736.

737.     Interaction interaction =

738.         client.interactions.create(CreateInteractionRequestBody.of
            (params)).interaction().get();

739.

740.         System.out.println(interaction.outputText().orElse(""));

741.

742.     ### REST

743.

744.     curl
        "https://generativelanguage.googleapis.com/v1beta/inter
        actions" \

745.         -H 'Content-Type: application/json' \

746.         -H "x-goog-api-key: YOUR_API_KEY" \

747.         -X POST \

748.         -d '{

749.             "model": "gemini-3.8-flash",
```

Quiz:

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750. "input": "Explain how AI works in a few words"
751. }'
- 752.
753. ## Security and secret management
- 754.
755. Treat your Gemini API key like a password. If
 compromised, others can consume
756. your project's quota, incur unexpected billing charges,
 and access private
757. resources.
- 758.
759. ### Critical security rules
- 760.
761. - **Keep keys confidential**: Never check API keys into
 source control systems like Git.
762. - **Never expose keys client-side in production**: Do
 not hardcode API keys directly in web or mobile apps. Keys
 compiled in client-side code can be extracted by users. To
 secure client-side apps, run a backend proxy server to
 make the actual API calls.
- 763.

Quiz:

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        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

764. **### Secret management best practices**

765.

766. - ****Environment variables****: Read keys from environment variables rather than configuration files.

767. - ****Secret Manager**** : For production, store your keys in a secure secret store such as [Google Cloud Secret Manager](https://cloud.google.com/secret-manager).

768. - ****Billing alerts****: Set up billing alerts in the Google Cloud Console to notify you if usage or costs spike.

769.

770. **### Leak response checklist**

771.

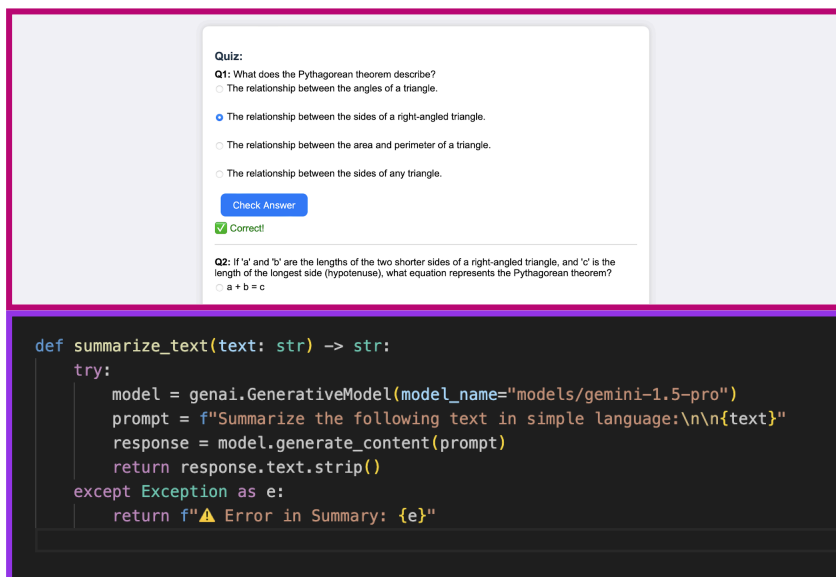
772. If you suspect your API key has been leaked:

773.

774. 1. ****Generate a new key****: Create a replacement key in Google AI Studio or the Cloud Console.

775. 2. ****Update your application****: Deploy your code using the new key.

776. 3. ****Disable or delete the compromised key****: Disable the leaked key in the Cloud Console once the new key is



verified. Do not delete the old key until the new key is fully active to avoid application downtime.

777. 4. ****Audit usage****: Check billing logs and API usage in the Google Cloud Console to identify unauthorized activity.

778.

779. **## Restricting and securing your keys**

780.

781. Adding restrictions to your API keys minimizes the potential damage if a key is

782. compromised.

783.

784. **### Apply request origin restrictions**

785.

786. Origin restrictions limit which IP addresses, websites, or applications can use

787. your key.

788.

789. 1. Go to the [Google Cloud Console Credentials page](https://console.cloud.google.com/apis/credentials).

Quiz:

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```

790. 2. Select your project, and click the name of the API key you want to restrict.
791. 3. Under ****Application restrictions****, select ****IP addresses**** (or the appropriate restriction type for your environment).
792. 4. Specify the allowed IP addresses or ranges, then click ****Save****.
- 793.
794. **### Securing unrestricted standard API keys**
- 795.
796. To continue using the Gemini API, you must secure any unrestricted keys.
- 797.
798. **#### Method A: Restrict the key to the Gemini API only (AI Studio)**
- 799.
800. If you only use the key for the Gemini API, secure it directly in AI Studio:
- 801.

Quiz:

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    except Exception as e:
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```

802. 1. On the ****API Keys**** page in [Google AI Studio](https://aistudio.google.com/api-keys), locate keys marked with the ****Unrestricted**** label.
803. 2. Hover over the label and click ****Add restrictions**** in the dialog.
804. 3. Select ****Restrict to Gemini API only****.
805. 4. Click ****Restrict key**** to confirm.
- 806.
807. > [!NOTE]
808. > ****Note:**** To restrict an API key, you must have the ``apikey.keys.update`` permission on the associated Google Cloud project. This permission is included in roles like ****API Keys Admin**** or ****Editor****.
- 809.
810. ##### Method B: Restrict the key for other services (Google Cloud Console)
- 811.
812. If the key is shared with other Google APIs (not recommended), restrict it in
813. the Cloud Console. ****Note:** Gemini API requests using this key will fail after

Quiz:

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

814. these restrictions are applied.**
- 815.
816. 1. Visit the [Google Cloud Console Credentials page](https://console.cloud.google.com/apis/credentials).
817. 2. Select the project and the API key.
818. 3. Under ****API restrictions**** , use the ****Select API restrictions**** drop-down to select the APIs you want this key to access. Do not select the ****Generative**
819. Language API**.
820. 4. Click ****Save****. Create a separate, restricted key in AI Studio to continue using the Gemini API.
- 821.
822. **### Blocked dormant keys**
- 823.
824. Starting May 7, 2026, the Gemini API blocks unrestricted API keys that have
825. been dormant for an extended period. These keys show a ****Blocked**** tag in AI
826. Studio. You must generate a new key or use an existing restricted key to
827. continue.

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```

828.

829. **## Migrate to an auth key**

830.

831. Follow these steps to create a new auth API key and update your applications:

832.

833. 1. Go to the [AI Studio API Keys page](https://aistudio.google.com/api-keys).

834. 2. Check the ****Key Type**** column to identify any keys listed as ****Standard****.

835. 3. Click ****Create API key**** to generate a new key. All new keys created in AI Studio are automatically created as auth keys.

836. 4. Copy the new auth API key.

837. 5. Update your application code, environment variables, and any deployment configurations to use the new auth API key.

838. 6. Test your application to confirm it works correctly with the new key.

839. 7. Once verified, delete or revoke your old traffic key to prevent misuse.

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        return response.text.strip()
    except Exception as e:
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```

840.

841. > [!NOTE]

842. > **Note:** If you don't see your keys, you may need to import your Cloud projects into AI Studio. Refer to the [import projects](https://ai.google.dev/gemini-api/docs/api-key#import-projects) section for instructions.

843.

844. ## Limitations

845.

846. Google AI Studio imposes the following project and key management limitations:

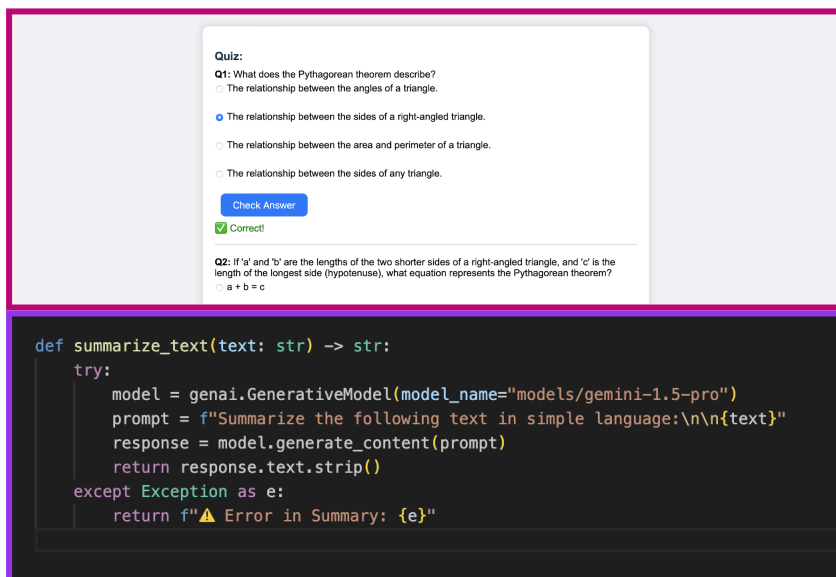
847.

848. - You can create a maximum of 10 projects at a time from the Google AI Studio **Projects** page.

849. - The **API keys** and **Projects** pages display a maximum of 100 keys and 50 projects.

850. - Only API keys that are unrestricted or restricted specifically to the Generative Language API (Gemini API) are displayed.

851.



852. For advanced project management or to modify keys with other restrictions, use
853. the [Google Cloud Console credentials page](https://console.cloud.google.com/apis/credentials).
- 854.
- 855.
856. curl
"https://generativelanguage.googleapis.com/v1beta/interactions" \
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864. ¶
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866. ~~¶~~

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867. ~~Treat your Gemini API key like a password. If compromised, others can consume~~¶¶
868. ~~your project's quota, incur unexpected billing charges, and access private~~¶¶
869. ~~resources.~~¶¶
870. ¶¶
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872. ¶¶
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892. ~~Language API~~.

893. ~~your application code, environment~~

894. ~~Official Documentation: Google AI for Developers~~

895. ~~Installation Guide: Tom's Hardware~~

- sooo good
-
- Popular Tutorial: [Python Programming Tutorial - Full Course for Beginners](#)
- Installation Steps:

1. Download the latest Python 3.10+ installer for your operating system from the official website.
2. Run the installer and ensure you check the box that says "Add Python to PATH".
3. Follow the installation prompts to complete the setup.
4. Verify the installation by opening a terminal or command prompt and typing `python --version`.

2. FastAPI Framework –

- Official Documentation: [FastAPI](#)
- User Guide: [FastAPI](#)
- Popular Tutorial: [FastAPI Crash Course](#)

3. HTML & CSS – Basic templating used in /templates and /static

4. Google Gemini API Key

- Official Documentation: [Google AI for Developers](#)
- Setup Guide: [GeeksforGeeks](#)

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- Popular Tutorial: [How to Use Google Gemini API Key](#)
- Setup Steps:
 1. Visit the Google AI Studio and sign in with your Google account.
 2. Create a new project and enable the Gemini API.
 3. Generate an API key and securely store it.

5. Uvicorn (ASGI Server)

- Official Documentation: [PyPI](#)
- Installation Steps:.....
 1. Install Uvicorn using pip: `pip install uvicorn`

6. Jinja2 (HTML Templating Engine)

- Official Documentation: [Jinja2 Documentation](#)
- Installation Steps:
 1. Install Jinja2 using pip: `pip install jinja2`

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Project Workflow

MILESTONE 1: Model Selection and Architecture

Activity 1.1: Select AI Models

- Gemini 1.5 Pro (via API):**
 Used for: Q&A, summarization, quiz generation, learning paths
 Benefits: Advanced reasoning, structured outputs, and cloud inference
- LaMini-Flan-T5-783M (local):**
 Used for: Concept explanation
 Benefits: Instruction-tuned, lightweight, CPU-compatible

Folder Architecture:

EduGenie/		
1.	main.py	# FastAPI app
2.	explanation_module.py	# Concept explanation logic

Quiz: Q1: What does the Pythagorean theorem describe? ☐ The relationship between the angles of a triangle. ☒ The relationship between the sides of a right-angled triangle. ☐ The relationship between the area and perimeter of a triangle. ☐ The relationship between the sides of any triangle. Check Answer ✔ Correct! Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem? ☐ $a + b = c$	
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3.	qna.py # Question answering
4.	quiz_module.py # Quiz generation
5.	summary_module.py # Summarization
6.	learning_path.py # Learning recommendations
7.	templates/index.html # HTML frontend
8.	static/style.css # Styling
9.	requirements.txt # Python dependencies

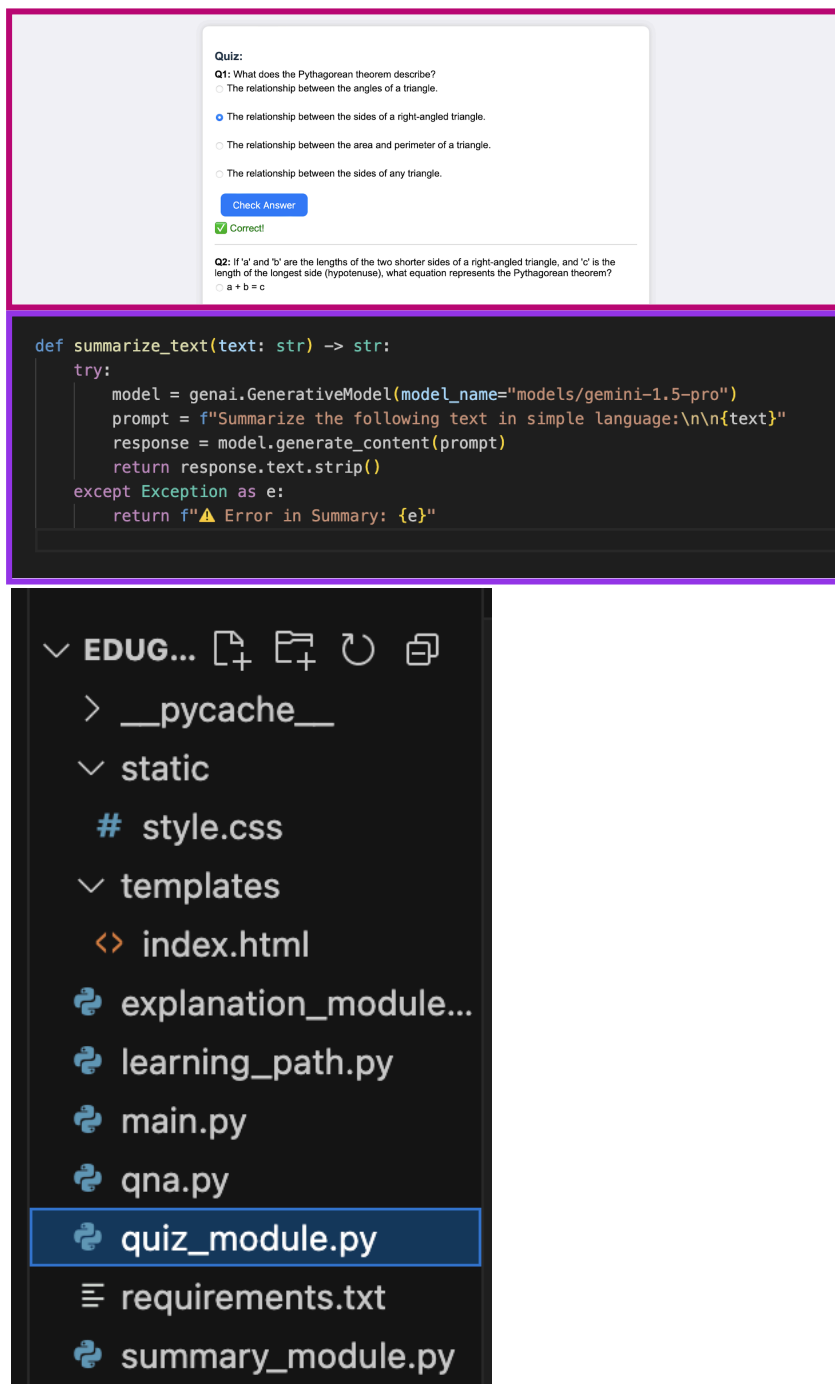


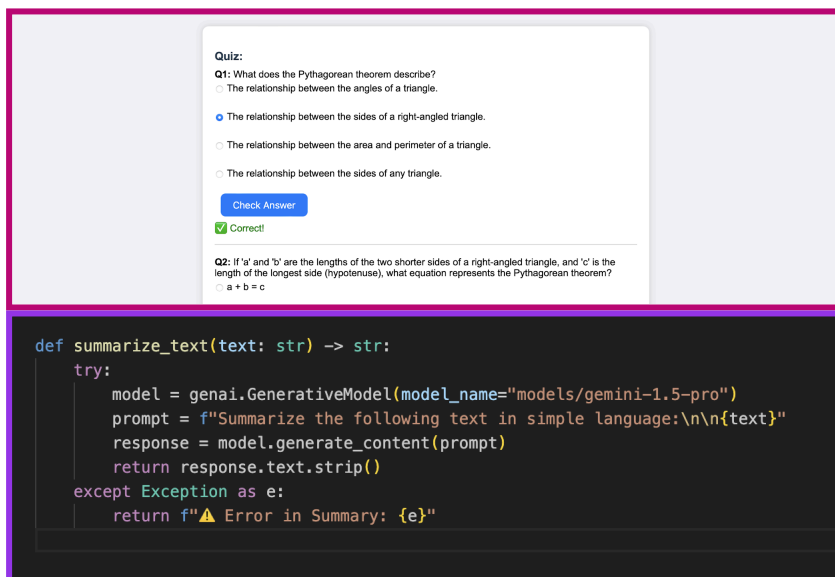
FIG. Showing the Folder Architecture

MILESTONE 2: Core Functionalities Development

Activity 2.1. : Module Implementation

1. Explanation Module

EduGenie utilizes the LaMini-Flan-T5 model, a lightweight yet powerful generative AI, to deliver educational content in a



simplified and highly readable manner. This model is specifically fine-tuned to provide concise, context-aware responses that break down complex topics into easily understandable language. By focusing on clarity and brevity, EduGenie ensures that learners with minimal background knowledge or technical experience can grasp essential concepts without feeling overwhelmed. This makes it particularly valuable for beginners, school students, or self-learners seeking straightforward explanations. The integration of LaMini-Flan-T5 empowers EduGenie to act as a reliable and accessible study companion for foundational learning across subjects.

```

Explanation_module.py
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
import torch

# Load the improved model for explanations
explain_tokenizer = AutoTokenizer.from_pretrained("MBZUAI/LaMini-Flan-T5-783M")
explain_model = AutoModelForSeq2SeqLM.from_pretrained("MBZUAI/LaMini-Flan-T5-783M")

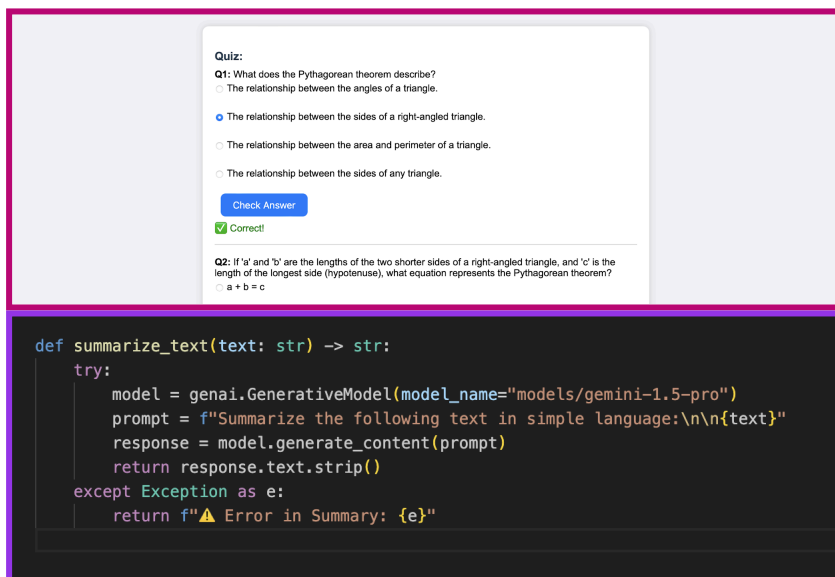
def explain_topic(topic: str) -> str:
    input_text = f"Explain the concept of '{topic}' in a simple and clear way for a school student."
    inputs = explain_tokenizer(input_text, return_tensors="pt")

    outputs = explain_model.generate(
        **inputs,
        max_new_tokens=150,
        temperature=0.7,
        top_k=50,
        top_p=0.95,
        do_sample=True
    )

    explanation = explain_tokenizer.decode(outputs[0], skip_special_tokens=True)
    return explanation
  
```

Fig. Explanation module logic

2. QnA Module



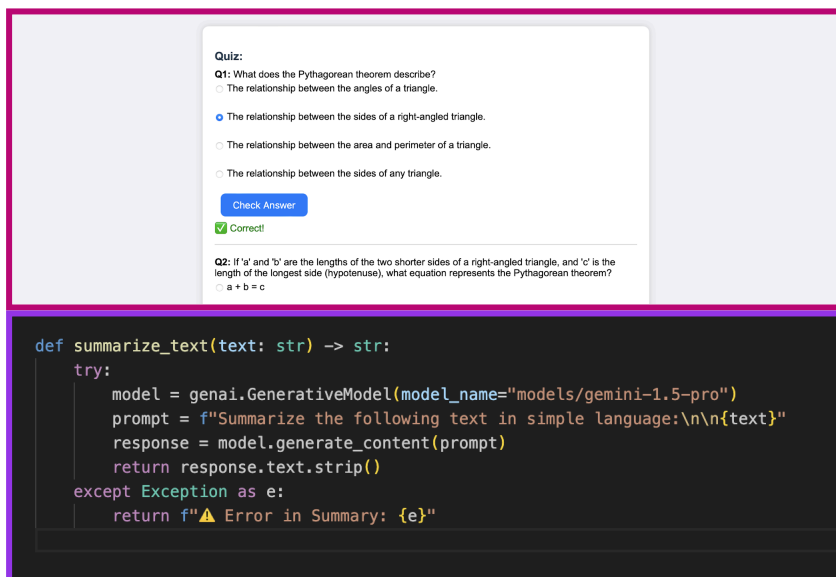
EduGenie is powered by Gemini 1.5 Pro, enabling it to handle a wide range of general knowledge and academic question-answering tasks with precision. Its advanced understanding and contextual capabilities make it an effective tool for learners seeking accurate, AI-driven assistance across diverse educational topics and subjects.



Fig. Q/A Module

3. Quiz Module

- EduGenie generates three multiple-choice questions (MCQs) from a given passage, each containing four carefully crafted options. It



utilizes the Gemini model to comprehend the context and meaning of the passage, ensuring that the questions are relevant and the distractors are plausible. The final output is structured in JSON format for easy integration.

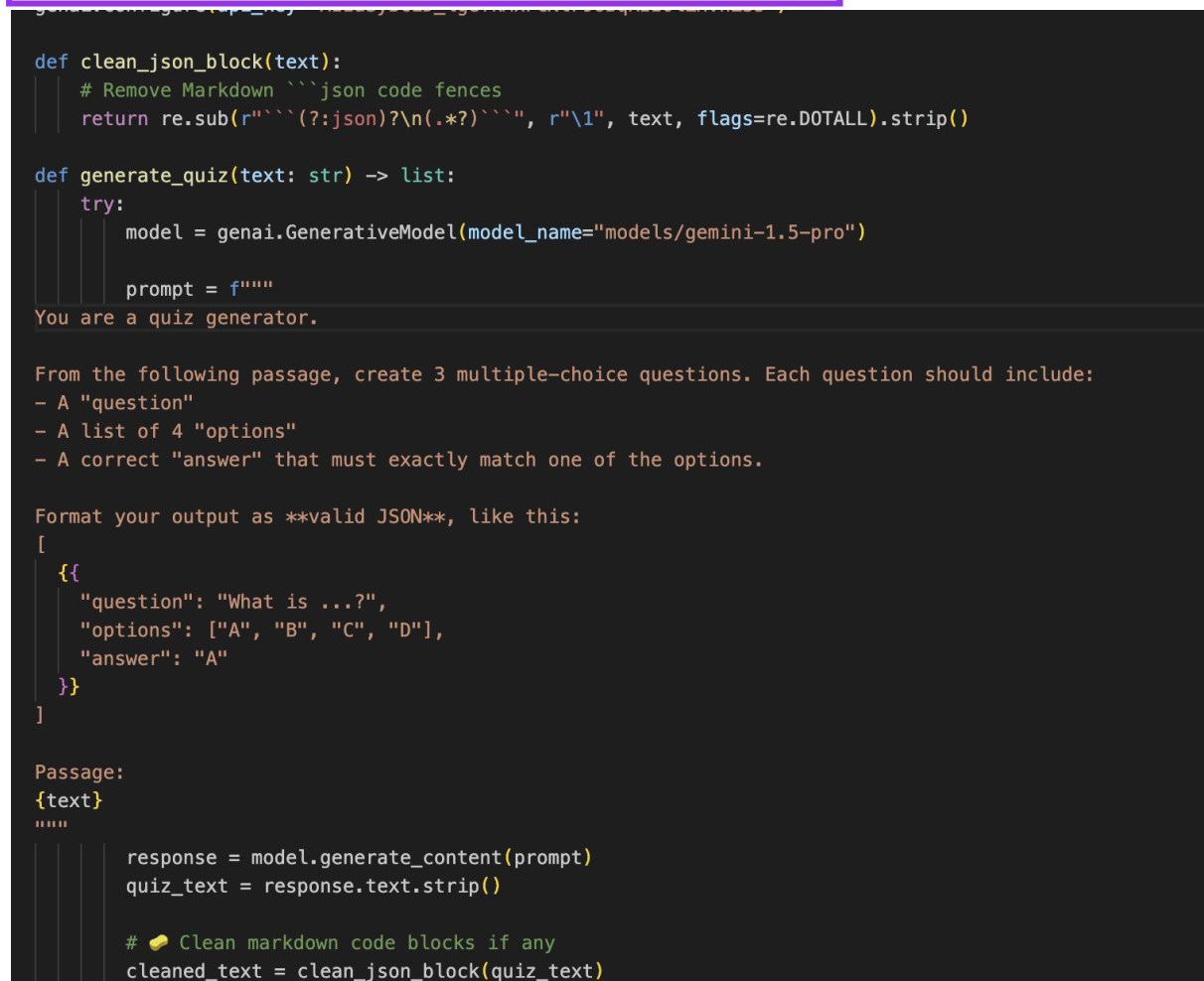
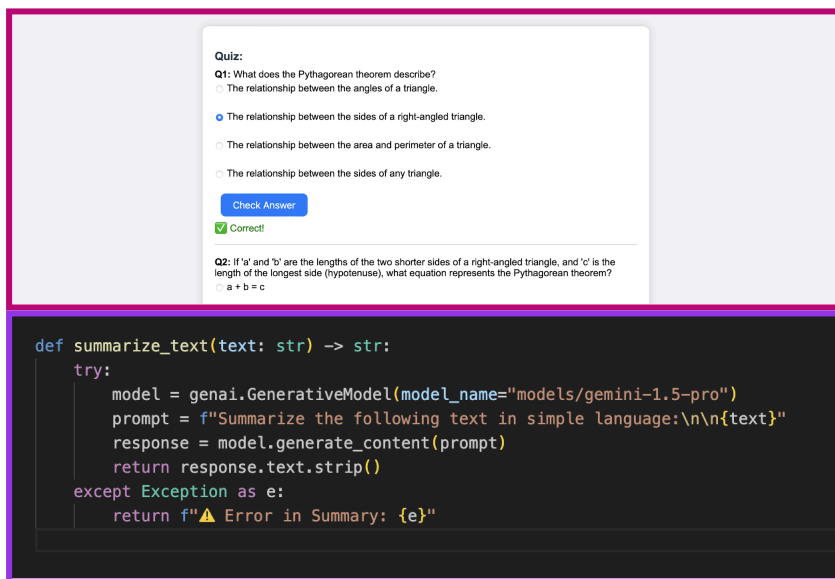
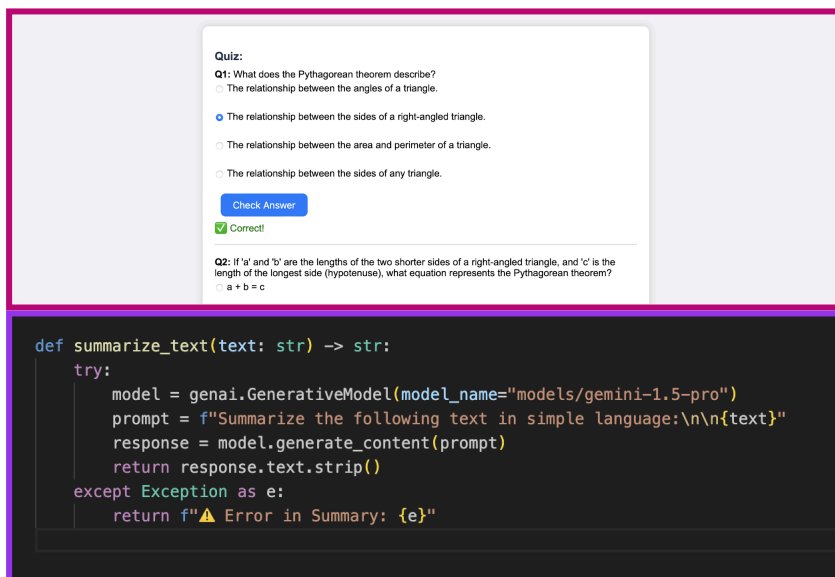


Fig. Quiz Module Logic

It sends a structured prompt to the model, expecting a valid JSON response with each question containing four options and a correct answer. To ensure proper parsing, it cleans any Markdown code



blocks using the `clean_json_blockfunction`. The cleaned response is then parsed into a Python list. If an error occurs during generation or parsing, it returns a detailed error message for debugging.

4. Summary Module

This feature leverages Gemini's generative capabilities to summarize long paragraphs into concise, easy-to-understand versions, making it ideal for quick revision. It ensures the core information is retained while eliminating redundancy, helping learners absorb key points efficiently without losing clarity or context.

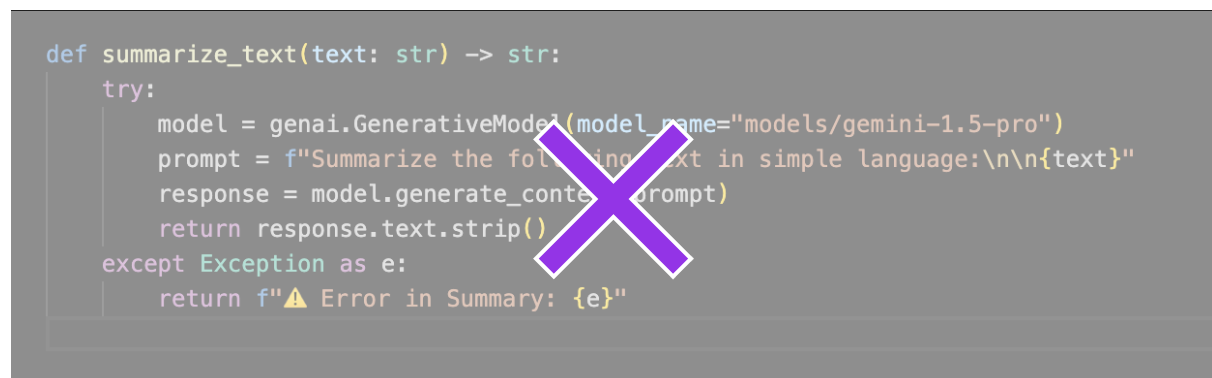
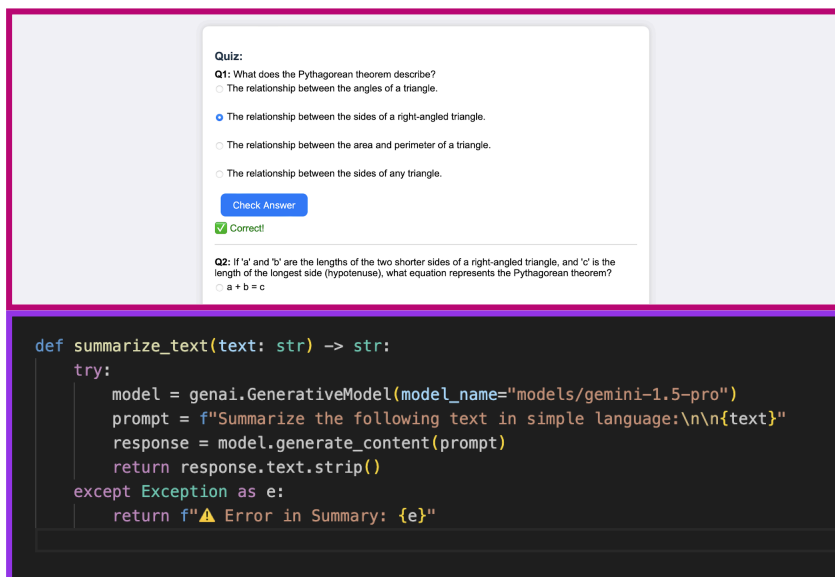


Fig. Summary Module

5. Learning Path Module

The `get_learning_recommendations` function uses Google Gemini to generate a personalized, structured learning path for any given topic. By crafting a detailed prompt, it instructs the AI to suggest



beginner to advanced concepts, organized by difficulty, and supported with useful resources such as videos, articles, or books. It ensures adaptability to the learner's level. The function safely handles API responses and errors, providing either the generated text or a helpful error message if the response is invalid or incomplete.

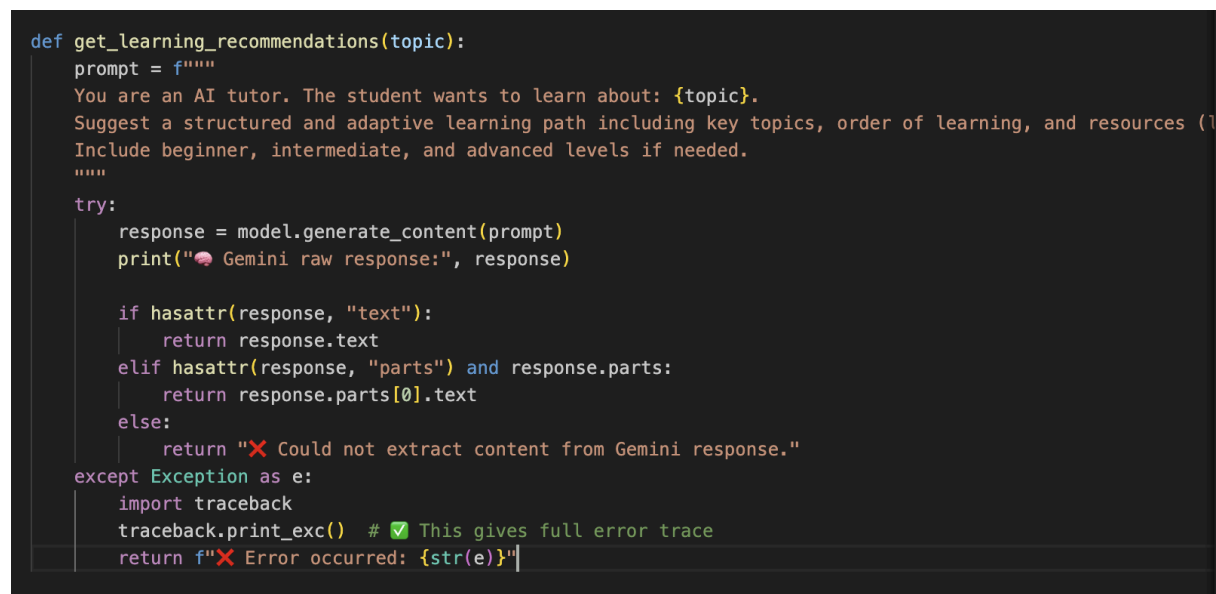
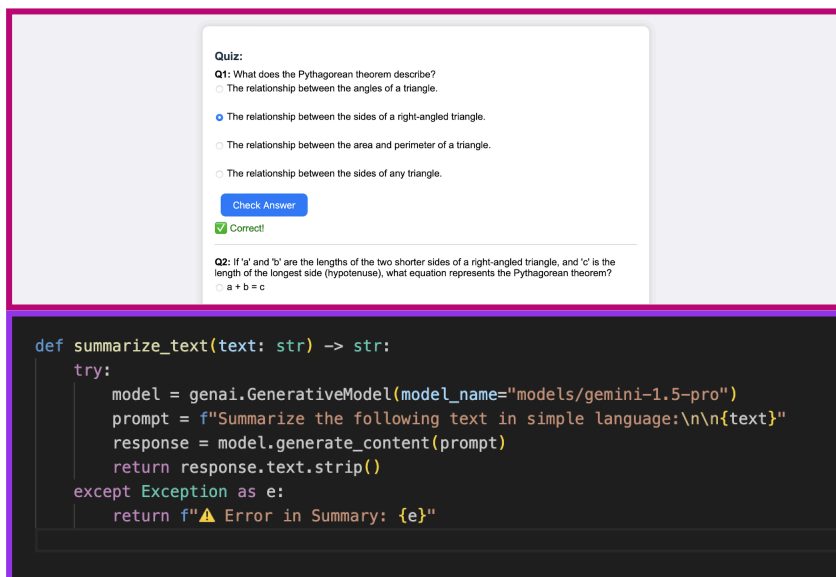


Fig. Learning Path Module

Activity 2.2 : Backend API with FastAPI

- Defined RESTful endpoints for each module:
 - /qa



- /explain
 - /quiz
 - /summarize
 - /learn/recommendations
- Connected each endpoint to a respective module logic

```

main.py
40
41 # Q&A - GET API using Gemini
42 @app.get("/qa")
43 async def answer_question(question: str = Query(...)):
44     answer = answer_question_with_gemini(question)
45     return {"answer": answer}
46
47 # Explanation - POST API
48 @app.post("/explain/")
49 async def explain_api(request: Request):
50     data = await request.json()
51     topic = data.get("topic")
52     if not topic:
53         return JSONResponse(content={"error": "Please provide a topic."}, status_code=400)
54     explanation = explain_topic(topic)
55     return {"topic": topic, "explanation": explanation}
56
57 # Summarization - POST API
58 @app.post("/summarize/")
59 async def summarize_api(request: Request):
60     data = await request.json()
61     text = data.get("text")
62     if not text:
63         return JSONResponse(content={"error": "Please provide text to summarize."}, status_code=400)
64     summary = summarize_text(text)
65     return {"summary": summary}
66

```

Fig. Endpoints of API

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

Check Answer

✔ **Correct!**

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

☐ $a + b = c$

```

def summarize_text(text: str) -> str:
    try:
        model = genai.GenerativeModel(model_name="models/gemini-1.5-pro")
        prompt = f"Summarize the following text in simple language:\n\n{text}"
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
  
```

```

66
67 # Quiz Generation - POST API
68 @app.post("/quiz")
69 async def quiz_api(request: Request):
70     data = await request.json()
71     text = data.get("text")
72     if not text:
73         return JSONResponse(content={"error": "Please provide text for quiz."}, status_code=400)
74     quiz = generate_quiz(text)
75     print("Generated quiz:", quiz) # ✔ DEBUG
76     return JSONResponse(content={"quiz": quiz})
77
78 # Learning Recommendations - GET API
79 @app.get("/learn/recommendations")
80 async def learning_recommendation_api(topic: str = Query(...)):
81     recommendation = get_learning_recommendations(topic)
82     return {"topic": topic, "recommendation": recommendation}
83
84
  
```

Fig. Endpoints of API

MILESTONE 3: Frontend Development

Activity 3.1: Build Web Interface

- HTML Form (index.html):
 - Task dropdown (Explain, QnA, Quiz, Summary, Recommend Path)
 - Text area for user input
 - Submit button
- CSS Styling (style.css):
 - Responsive design
 - Styled buttons, input areas, and result container

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

Check Answer

✔ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

☐ $a + b = c$

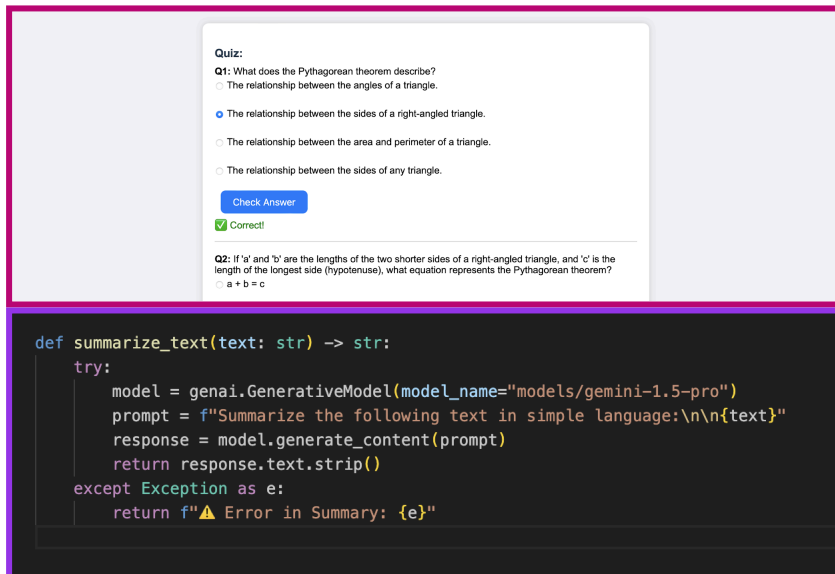
```
def summarize_text(text: str) -> str:
    try:
        model = genai.GenerativeModel(model_name="models/gemini-1.5-pro")
        prompt = f"Summarize the following text in simple language:\n\n{text}"
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

```
templates > <> index.html > ...
2   <html lang="en">
3   <head>
86  </head>
87  <body>
88      <h1>Welcome to EduGenie 🤖!</h1>
89      <p>Your personal AI tutor for learning support!</p>
90
91      <!-- Q&A -->
92      <form id="qaForm">
93          <label><strong>Ask EduGenie a Question:</strong></label><br><br>
94          <input type="text" id="question" placeholder="Why is the sky blue?" required>
95          <button type="submit">Get Answer</button>
96      </form>
97      <div id="qaResult" class="output"></div>
98
99      <!-- Explanation -->
100     <form id="explainForm">
101         <label><strong>Need an Explanation?</strong></label><br><br>
102         <input type="text" id="topic" placeholder="Photosynthesis" required>
103         <button type="submit">Explain</button>
104     </form>
105     <div id="explanationResult" class="output"></div>
106
107     <!-- Summary -->
108     <form id="summaryForm">
109         <label><strong>Summarize a Paragraph:</strong></label><br><br>
110         <input type="text" id="summaryText" placeholder="Paste long content to summarize" required>
111         <button type="submit">Summarize</button>
112     </form>
113     <div id="summaryResult" class="output"></div>
114
115     <!-- Quiz -->
116     <form id="quizForm">
117         <label><strong>Generate a Quiz:</strong></label><br><br>
118         <input type="text" id="quizText" placeholder="Solar System" required>
119         <button type="submit">Generate Quiz</button>
120     </form>
121     <div id="quizResult" class="output"></div>
```

Fig. Showing Frontend Intergration

Activity 3.2: Live Integration

- Form submission sends POST request to FastAPI backend



- Result appears below input box in real-time

MILESTONE 4: Deployment

- **Activity 4.1: Run Locally**
Run locally with `uvicorn main:app --reload`
Navigate to: <http://127.0.0.1:8000>

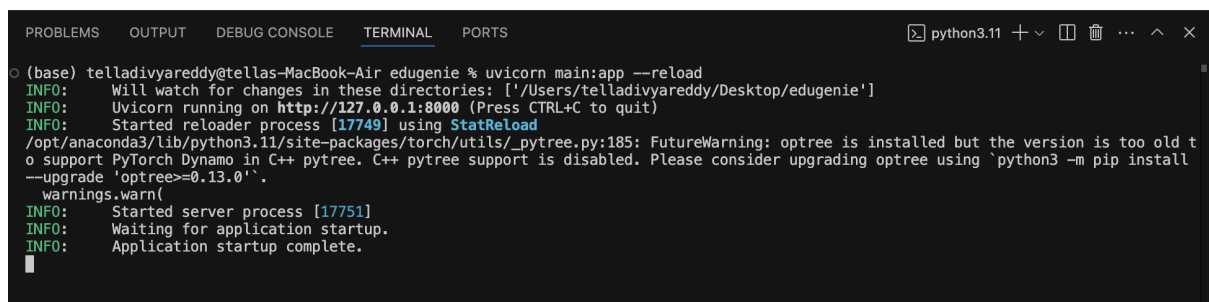


Fig. Showing the Application can be accessed

- **Activity 4.2: Functional Testing**
Ask a question
Get an explanation
Generate a quiz
Summarize content
Get a personalized learning plan

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.
☒ The relationship between the sides of a right-angled triangle.
☐ The relationship between the area and perimeter of a triangle.
☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ **Correct!**

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

☐ $a + b = c$
☐ $a^2 + b^2 = c^2$
☐ $a^2 + c^2 = b^2$
☐ $a^2 + b^2 = c$

```
def summarize_text(text: str) -> str:
    try:
        model = genai.GenerativeModel(model_name="models/gemini-1.5-pro")
        prompt = f"Summarize the following text in simple language:\n\n{text}"
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

Welcome to EduGenie 🧠✨

Your personal AI tutor for learning support!

Ask EduGenie a Question:

[Get Answer](#)

Need an Explanation?

[Explain](#)

Summarize a Paragraph:

[Summarize](#)

Fig. EDUGENIE

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ **Correct!**

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

Summarize a Paragraph:

Paste long content to summarize [Summarize](#)

Generate a Quiz:

Solar System [Generate Quiz](#)

Get Learning Recommendations:

e.g. Linear Regression [Get Recommendations](#)

Fig. EDUGENIE

Exploring EduGenie:

- Asking questions:

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☒ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ Correct!

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

Welcome to EduGenie 🧠✨

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Ask EduGenie a Question:

which is the largest ocean?

[Get Answer](#)

Answer:

The Pacific Ocean is the largest ocean on Earth.

b. Explanation of any topic:

Need an Explanation?

quantum computing

[Explain](#)

Explanation:

Quantum computing is a type of computing that uses the principles of quantum mechanics to perform certain calculations much faster than traditional computers. It allows computers to perform certain calculations much faster than traditional computers because it uses tiny particles called qubits that can exist in multiple states at the same time. This allows for new kinds of calculations and solving problems that are not possible with traditional computers.

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ **Correct!**

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

Need an Explanation?

Binary Search Algorithm [Explain](#)

Explanation:

The binary search algorithm is a way to find the index of a target value in a sorted list of numbers. It's like searching for a needle in a haystack. You need to find the number between 0 and 1, and then go back to the beginning of the list to find the target value.

c. Summarising long paragraphs:

Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century, mark [Summarize](#)

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

d. Generating quizzes:

It generates three questions with 4 options each. It corrects the answer if chosen wrong.

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☒ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

[Check Answer](#)

✓ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

- ☐ $a + b = c$

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

Generate a Quiz:

Pythagoras theorem

[Generate Quiz](#)

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

[Check Answer](#)

Quiz:

Q1: What does the Pythagorean theorem describe?

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- ☒ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

[Check Answer](#)

✓ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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[Check Answer](#)

✔ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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        prompt = f"Summarize the following text in simple language:\n\n{text}"
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

- ☐ $a + b = c$
- ☐ $a^2 + b^2 = c^2$
- ☒ $a^2 - b^2 = c^2$
- ☐ $2a + 2b = 2c$

[Check Answer](#)

✗ Incorrect. Correct answer: $a^2 + b^2 = c^2$

Q3: Which type of triangle does the Pythagorean theorem apply to?

- ☐ Equilateral triangles
- ☐ Isosceles triangles
- ☒ Right-angled triangles
- ☐ All types of triangles

[Check Answer](#)

✔ Correct!

- e. Guiding with learning recommendations from beginners level to advanced level

Quiz:**Q1:** What does the Pythagorean theorem describe?

- ☒ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

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        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in Summary: {e}"
```

Get Learning Recommendations:

SQL

[Get Recommendations](#)**Learning Recommendations for "SQL":****## SQL Learning Path: From Zero to Hero**

This learning path is structured to progressively introduce SQL concepts, starting from the basics and gradually advancing to more complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

****I. Beginner Level: Building a Foundation********(Estimated Time: 1-2 weeks)******* **Key Topics:****

- * What is a Database and SQL? (Relational Model, DBMS)
- * Basic Syntax (SELECT, FROM, WHERE)
- * Data Types (INT, VARCHAR, DATE, etc.)
- * Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)
- * Ordering Results (ORDER BY)
- * Limiting Results (LIMIT/OFFSET or TOP)
- * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☒ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

- ☒ $a^2 + b^2 = c^2$

```
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        return response.text.strip()
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        return f"⚠ Error in Summary: {e}"
```

complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

****I. Beginner Level: Building a Foundation****

****(Estimated Time: 1-2 weeks)****

*** **Key Topics:****

- * What is a Database and SQL? (Relational Model, DBMS)
- * Basic Syntax (SELECT, FROM, WHERE)
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- * Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)
- * Ordering Results (ORDER BY)
- * Limiting Results (LIMIT/OFFSET or TOP)
- * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
- * Introduction to Tables and Columns

*** **Resources:****

- * ****Interactive Tutorials:****
 - * SQLZoo: Excellent for hands-on practice with different SQL flavors.
 - * Khan Academy's SQL course: Offers a good introductory overview.
 - * Codecademy's Learn SQL: Interactive lessons with practical exercises.
- * ****Videos:****
 - * FreeCodeCamp's SQL Tutorial: Comprehensive and beginner-friendly.

****II. Intermediate Level: Working with Multiple Tables****

****(Estimated Time: 2-3 weeks)****

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☒ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

[Check Answer](#)

✔ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

- ☐ $a + b = c$

```
def summarize_text(text: str) -> str:
    try:
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```

****II. Intermediate Level: Working with Multiple Tables****

****(Estimated Time: 2-3 weeks)****

*** **Key Topics:****

- * Joining Tables (INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN)
- * Subqueries (Nested queries)
- * Grouping Data (GROUP BY)
- * Filtering Groups (HAVING)
- * Set Operations (UNION, INTERSECT, EXCEPT)
- * Working with Strings (string concatenation, substrings, LIKE operator)
- * Working with Dates and Times (date formatting, date arithmetic)
- * Views (Creating and using views)

*** **Resources:****

- * ****Books:****
 - * "SQL Queries for Mere Mortals" by Michael J. Hernandez: A great resource for understanding practical SQL querying.
- * ****Online Documentation:**** Refer to the documentation for your specific SQL database (e.g., MySQL, PostgreSQL, SQL Server).
- * ****Practice Platforms:**** LeetCode, HackerRank, and similar platforms offer SQL challenges to reinforce your skills.

****III. Advanced Level: Mastering Database Management****

****(Estimated Time: 3-4 weeks and beyond)****

*** **Key Topics:****

- * Stored Procedures and Functions
- * Triggers
- * Indexes and Performance Tuning
- * Transactions and Concurrency Control
- * Database Design (Normalization)

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

Check Answer

✔ **Correct!**

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

☐ $a + b = c$

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```
* **Books:**
* "SQL Performance Explained" by Markus Winand: In-depth
coverage of SQL performance optimization.
* "Database Internals" by Alex Petrov: Explains the inner
workings of database systems.
* **Advanced Online Courses:** Look for specialized courses on
database administration and performance tuning on platforms like Udemy
and Coursera.
* **Community Forums and Blogs:** Engage with the SQL community to
learn from experienced professionals and stay up-to-date with the latest
advancements.

**Adaptive Learning Tips:**

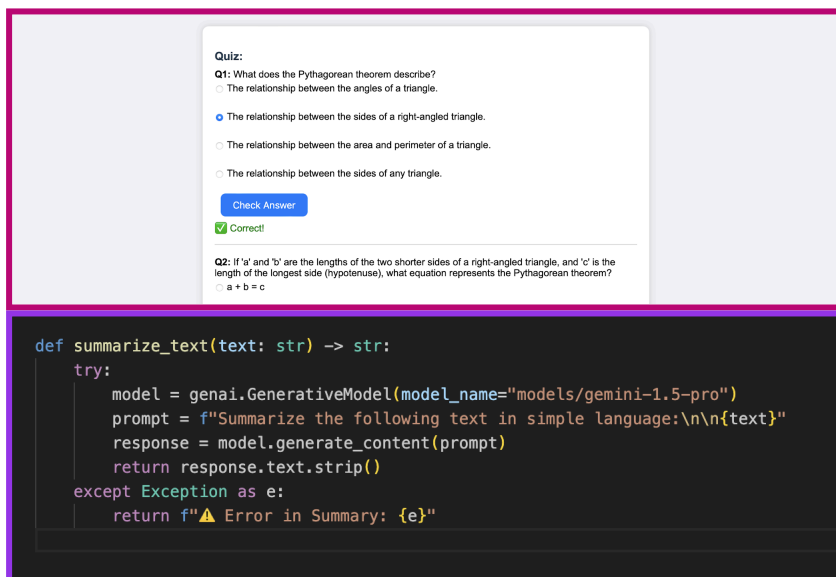
* **Start with the basics:** Don't rush into advanced topics before
mastering the fundamentals.
* **Practice regularly:** The more you practice writing SQL queries, the
better you'll become.
* **Use real-world datasets:** Working with real data will help you
understand how SQL is used in practical scenarios.
* **Focus on understanding the concepts:** Don't just memorize syntax;
understand the underlying logic behind SQL queries.
* **Break down complex problems:** Divide large problems into smaller,
manageable parts.
* **Don't be afraid to experiment:** Try different approaches and see
what works best.
* **Seek help when needed:** Don't hesitate to ask for assistance from
online communities, forums, or mentors.

This learning path provides a solid foundation for mastering SQL.
Remember to adapt it to your own learning style and goals, and enjoy the
journey of becoming proficient in this powerful language!
```

It provides along with details of resources from where to learn and stepwise guidance through-out.

Conclusion:

EduGenie has been developed as a robust and accessible AI-powered educational assistant that seamlessly integrates cloud-based intelligence with an intuitive, low-footprint web interface. Designed to support self-learners, educational institutions, and content platforms, EduGenie democratizes learning by simplifying complex concepts, providing instant question-and-answer interactions, generating personalized

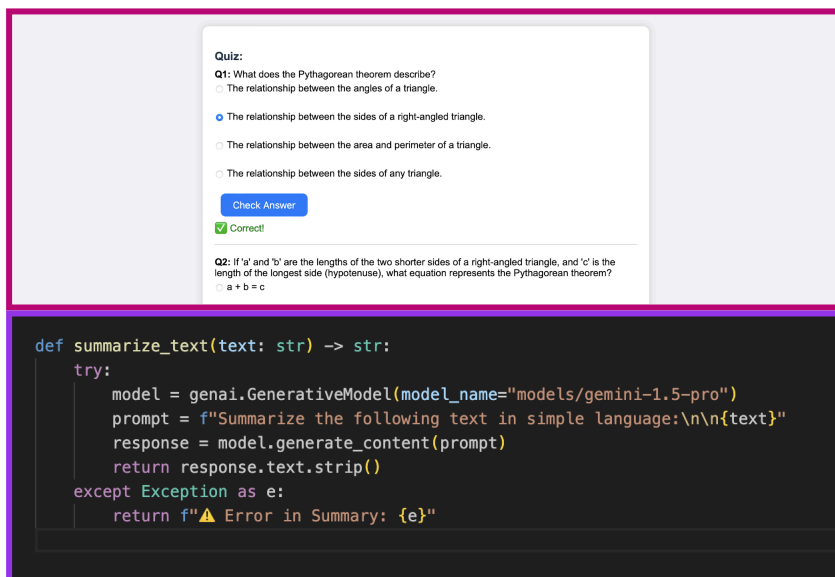


quizzes, offering detailed summaries, and tailoring smart learning paths based on individual needs. Its lightweight infrastructure ensures that even users with minimal hardware resources can access high-quality, personalized education without barriers.

Throughout the development of EduGenie, the project achieved key milestones in AI integration, user experience design, and content adaptability. Leveraging advanced generative AI, it transforms static learning into a dynamic and interactive process. The platform's simplicity and clarity cater to learners of all levels—especially those who may find traditional educational content overwhelming or inaccessible. The modularity of the platform further allows for easy future upgrades, while its cloud-based backend ensures scalability and data-driven intelligence.

However, the development journey came with its own set of challenges. Ensuring accuracy in AI-generated responses, maintaining performance across devices, and balancing personalization with general usability were significant hurdles. Overcoming these required continuous testing, fine-tuning AI prompt structures, optimizing the UI for clarity, and addressing data privacy concerns. Yet, these challenges provided valuable learning outcomes—especially in areas like prompt engineering, cloud integration, user-centric design, and scalable architecture.

Looking ahead, EduGenie holds immense potential for enhancement and expansion. One promising direction involves the integration of voice-based interaction, enabling users to learn hands-free using spoken commands and queries—a particularly useful feature for accessibility and multitasking. Multilingual support is another key frontier, helping bridge language barriers and reaching a broader global audience. A mobile application is also envisioned to make learning available anytime, anywhere, with offline features for uninterrupted access.



Furthermore, future iterations of EduGenie could incorporate progress tracking dashboards, allowing users to visualize their learning journey through metrics and insights. The addition of gamified elements such as badges, learning streaks, and challenge-based assessments can boost engagement and motivation. Adaptive learning paths powered by data analytics can guide users more intelligently through content based on their strengths and weaknesses.

From a collaboration perspective, EduGenie can evolve to support group study sessions, teacher or parent dashboards, and integration with Learning Management Systems (LMS) such as Moodle or Google Classroom. Input recognition for images and PDF files can expand how students interact with content, enabling snapshot-based doubt solving and resource summarization. Real-time sync and smart notifications will also improve continuity and interactivity.

In conclusion, EduGenie is not just an AI tool—it's a foundation for the future of personalized, inclusive, and intelligent education. By bridging gaps in access, comprehension, and adaptability, it redefines the learning experience for digital natives and underserved communities alike. As it continues to evolve, EduGenie is poised to become a comprehensive learning companion for the modern era—smart, scalable, and learner-first.

Submitted by:

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[Check Answer](#)

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Tella Divya Sree**Mentor: Siri****Date: 11/04/2025**